

Building Auotmation System

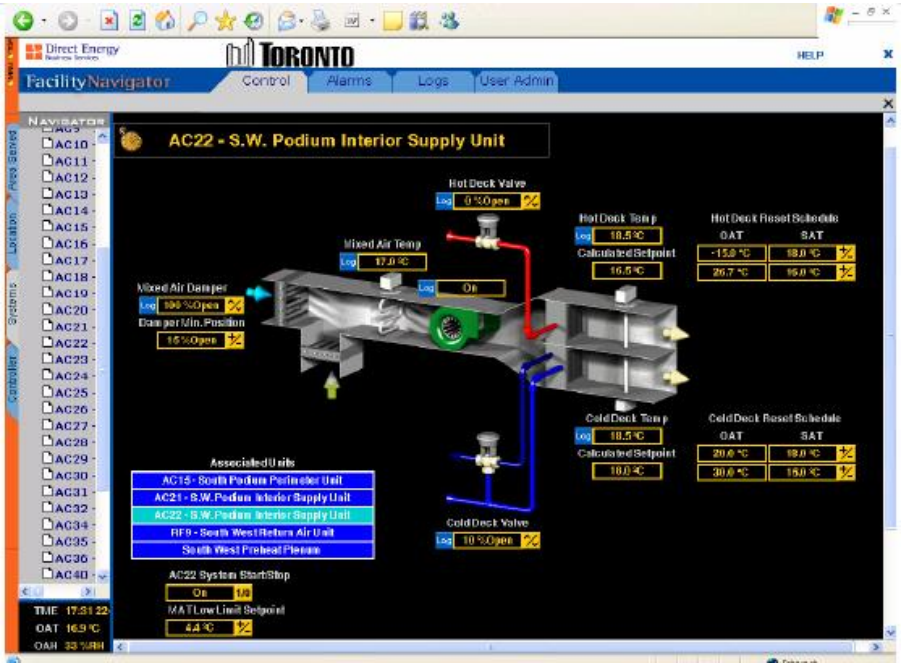
By

Samantha Gunawardana

What is a BMS?

- Different Names
  - BMS- Building Management System
  - BAS- Building Automation System
  - DDC - Direct Digital controls
  - CCMS – Central Control and Monitoring system

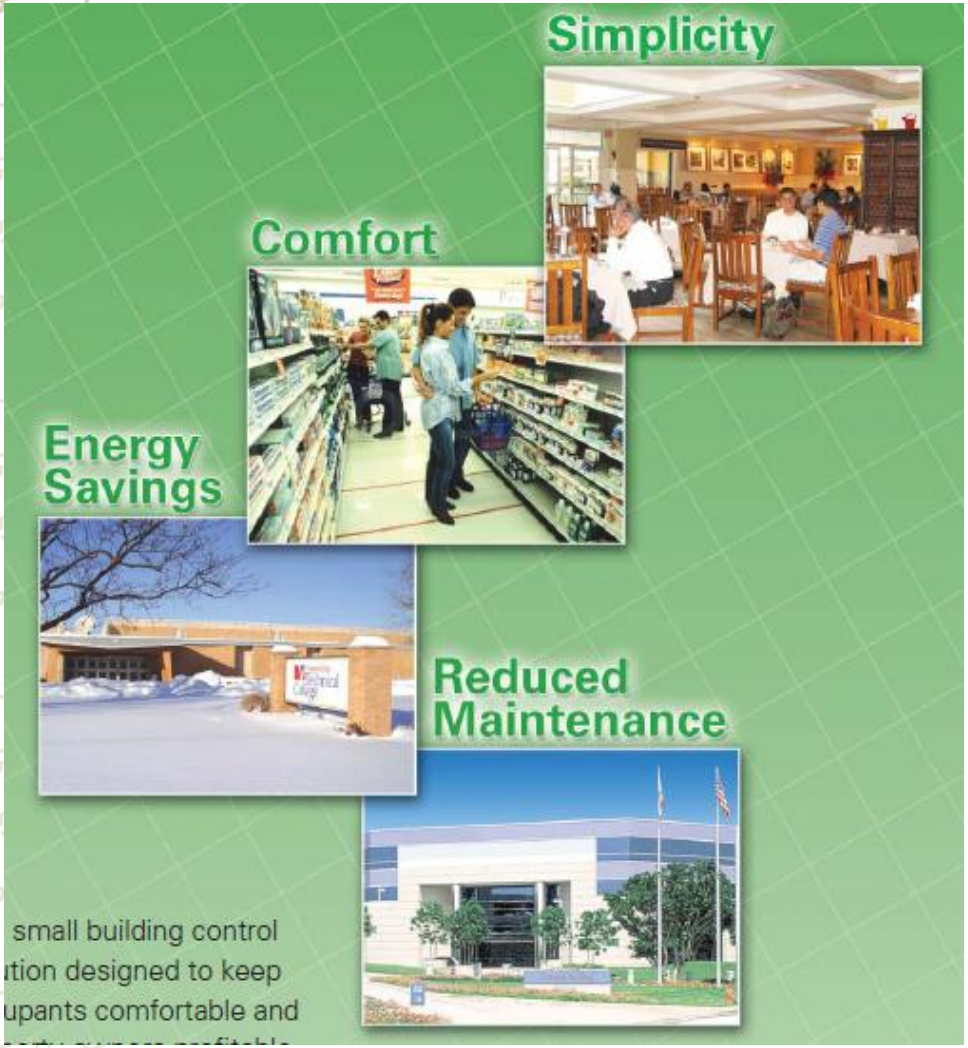




What is a BMS?

- Building Automation systems are networks of Microprocessor controls that
  - Controls the climate of the building
  - Supervise and control heating, ventilating and air conditioning equipment
  - Perform facility management(generate reports, graphs and annunciate alrams when there is a problem)
  - Perform energy management strategies to reduce operating and energy costs
  - Integrate building systems such as , security, fire alarm, lighting controls, etc

What is a BMS?


Simplicity
Comfort
Energy Savings
Reduced Maintenance
small building control
tion designed to keep
upants comfortable and


LEAVE ON HEAT OR AUTO (NOT COOL) UNTIL MAY
DO NOT TOUCH !! the thermostat
Make sure HOLD is not on leaving before office
DO NOT REMOVE COVER OR TOUCH BUTTONS

Direct Digital Controls

- DDC's meant for controlling air conditioning systems in commercial and Industrial buildings
- Direct digital control is a process in which a microprocessor controller constantly updates an internal database by monitoring information from a controlled environment and continuously produces the corrective output commands in response to the changing environment Controls the climate of the building

Direct Digital Controls

•Often gives the impression that the computer based systems are complicated.



•In reality microprocessor offer simple and organized way to build almost any electronic system.

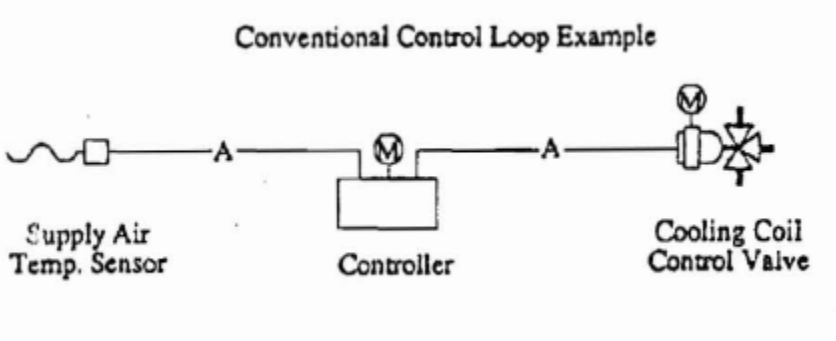
•All processing system including human system can be broken down into

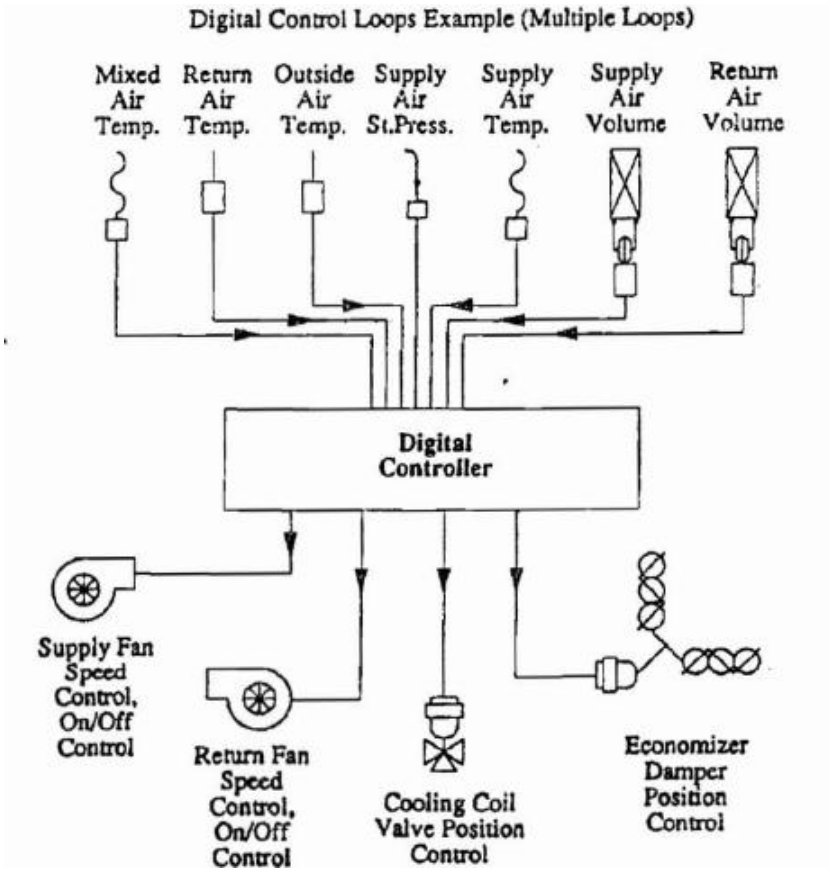
Sense → Decision → memory → action

DDC Vs Conventional

- Conventional controls are sometimes referred as Single loop controls(SLC)
- In Contrast Direct Digital Controls many control sequences simultaneously.

DDC Vs Conventional

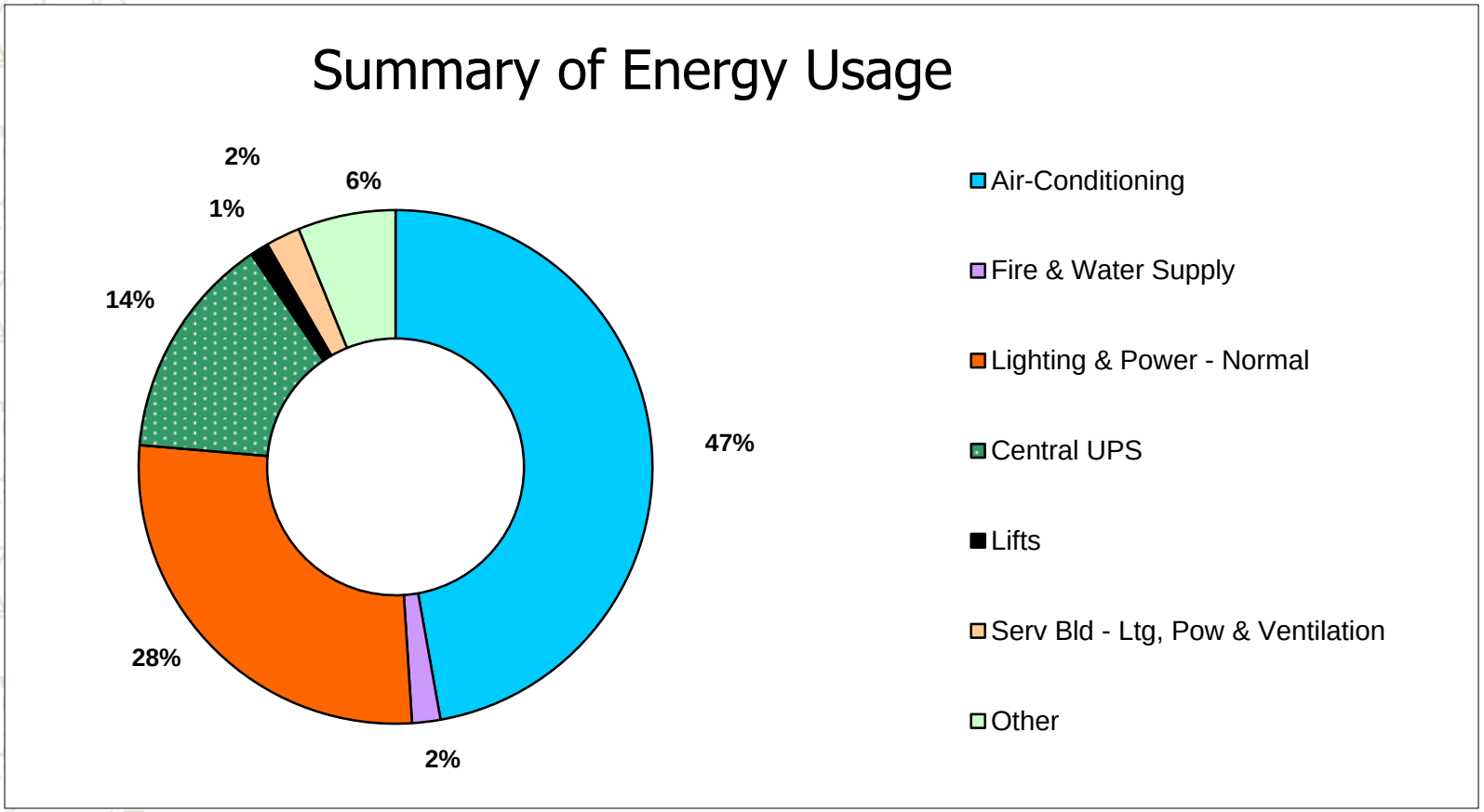




Operating cost

- Largest operating cost of commercial buildings are
  - Heating
  - Air conditioning(50VA/m2)
  - Lighting
  - Ventilation

Operating cost



Operating cost

- It is necessary to decrease energy cost without discomfort to the user
- Is it possible with a DDC?
  - Load shedding
  - Optimum start/stop
  - Reduction of speeds
  - Tight shut off of building(Lead to Sick building syndrome, Looses productivity and human health)

Operating cost

- In summary, effective and reliable building control systems are no longer an option
- It is a necessity.

BMS Designer

- Need to know about DDC
- Need to know the at least basics of what equipment to be controlled
- Related guidelines for those equipment
- Must be a expert on every system that he or she works with and seeks to control
- Should be familiar with computer systems
- Basics of electronics and electricity
- Knowledge of control theory

Development of Digital control

- In 1970, Due to rising energy cost
- Early machines, all processing at Central location, Local panel act as data gathering panels
- Very slow and Single failure cause the total system to collapse
- Less space for data storage

Development of Digital control

- Introduction of Personal computer, 1980 s, had a big impact in HVAC control industry
- DDC systems have travelled a hard road to popularity since their introduction in 1980s.
- The first generation controls were more error prone. Therefore the consulting engineering community tendered to favour the conventional controls.
- To make mattter worse
  - Difficult to program (No knowledge for operator) Not able to do changes

Development of Digital control

- Fortunately DDC systems have improved dramatically.
  - The current trend is smaller, modular, stand alone, at local level(Right at the HVAC equipment being controlled)
  - DDC hardware is more compact and reliable

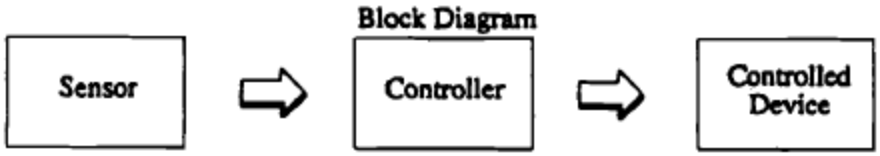
Development of Digital control

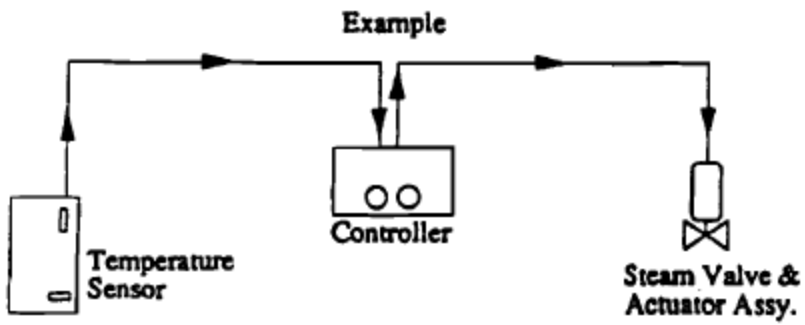


Fundamentals of Control system

- Purpose of automatic control system is to
  - Modulate the capacity of the equipment to satisfy the customer requirement
  - Monitor and control the safety of the operation











Fundamentals of Control system

- Technical terms
  - Controlled variable
  - Set point
  - Feedback
  - Closed loop control
  - Open loop control
  - Digital
  - Analog

Fundamentals of Control system

- Automatic control actions
  - Two position
  - Proportional
  - Proportional plus integral
  - Proportional plus integral plus derivative
  - Artificial intelligence

Inputs/Outputs and controllers

- Inputs and outputs of a DDC controller are the microprocessor's connections to the real world around it.

Input and Sensors

- Sensor reads the current value of the controlled variable
  - Sensors (Digital, Analog, Ex: Flow switch, Pt 100)
  - Transmitters and transducers(Transmitter amplifies, Transducer converts)
  - Sensor-Transmitter (In one package)
  - Sensor-controllers. (Eg: Thermostat)

Input and Sensors

- There are five specific type of sensors
  - Temp Sensors(Bimetal, Rod and tube, RTD, Thermistor, Thermocouple
  - Humidity sensors
  - Pressure sensors
  - Flow sensors
  - Special purpose sensor(watt, current, kw pulse,CO)

Input and Sensors

- Sensor selection consideration
  - Environment
  - Operating range
  - Sensor signal compatibility
  - Set point accuracy and repeatability
  - Response time (Fast response is not always desirable)

Controllers

- Controller is a device that collects information from the sensor, interprets this information, and send commands to controlled device to effect a corrective action.
- Types of controllers
  - Pneumatic
  - Electric
  - Electronic
  - Direct Digital
  - Hybrid

Controlled Devices

- Controlled devices are components that are controlled by mechanical or electrical means
- Common controlled devices are
  - Control Valves
  - Control Dampers
- Control Valves(Two position, Proportional)
  - Mech engineer to design the constant pressure drop across the valve
  - There are three general classifications of flow characteristics
    - Equal percentage
    - Linearity
    - Quick Opening

Controlled Devices

- Control Valves operator
  - To operate a motor control valve through its stroke a controlled device called Actuator is needed.
    - Electrical solenoid (Plunger is connected to valve stem)
    - Electric Motor actuators (Utilise a gear train)
      - Birdirectional
      - Spring return
      - Reversible

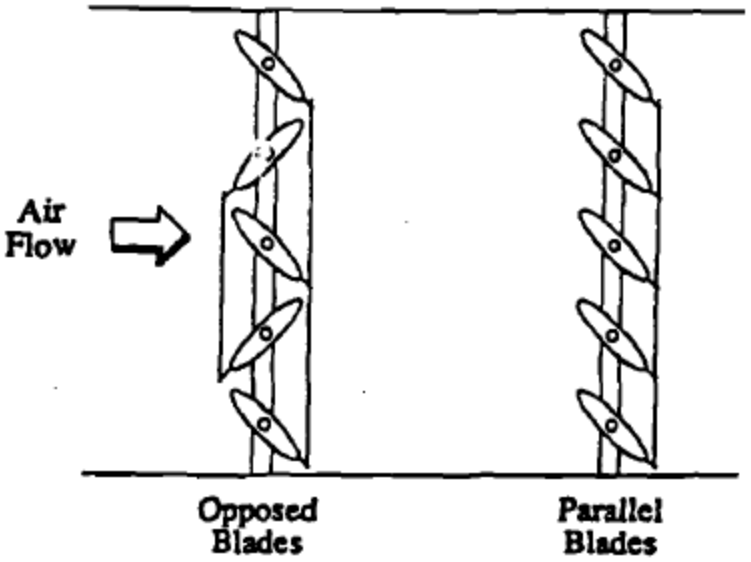


One type of actuator for many different operating modes

Controlled Devices

- Control Damper(A valve for air)
  - Two Position
  - Modulating control
- Damper actuator
  - Rotation shaft designed to link with damper
- Electrical Auxiliary devices





Controls System Architecture



Operator Interface

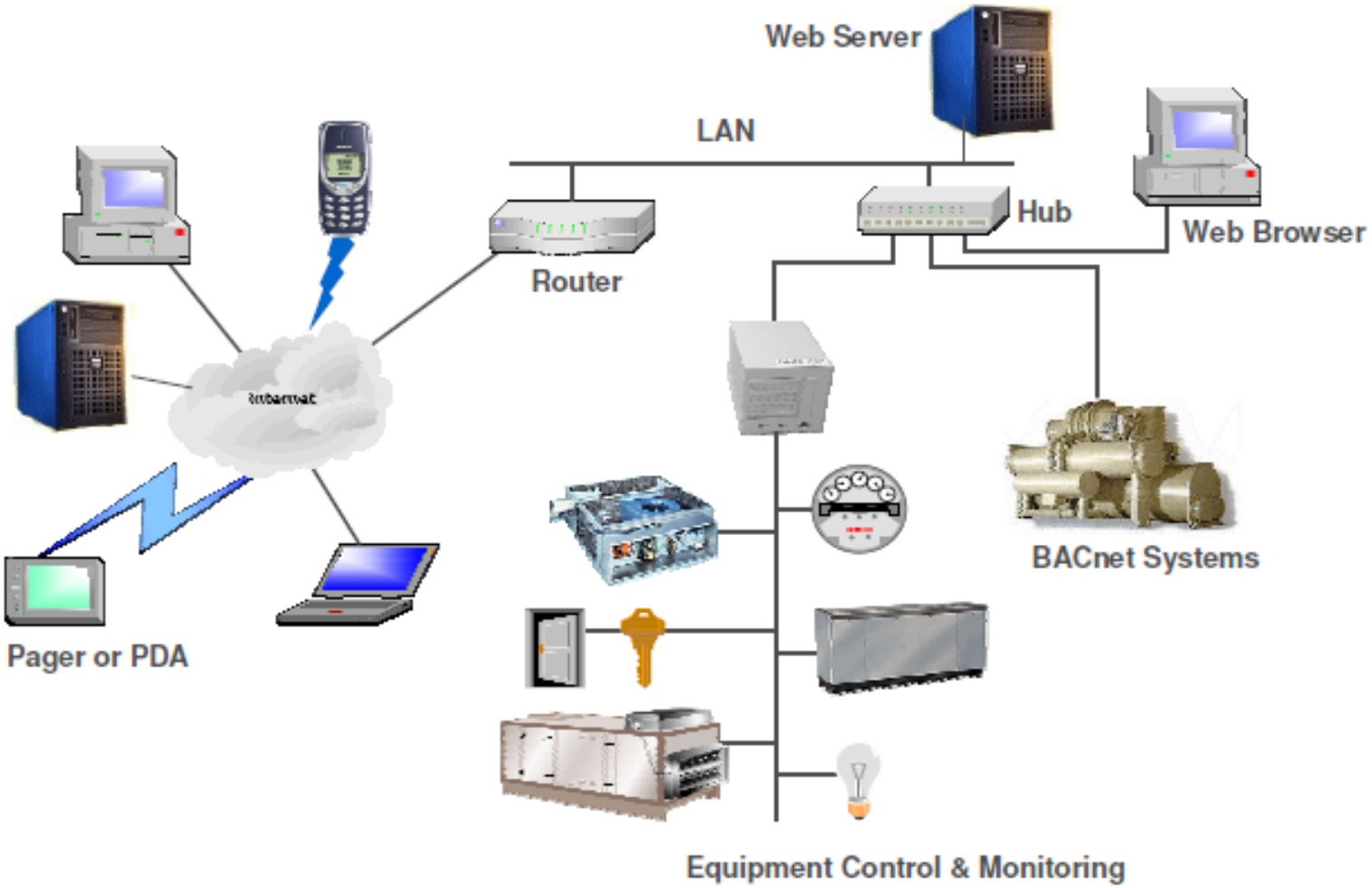


Building Control



Unit Control

Typical BAS Architecture



Direct Digital control Application strategies

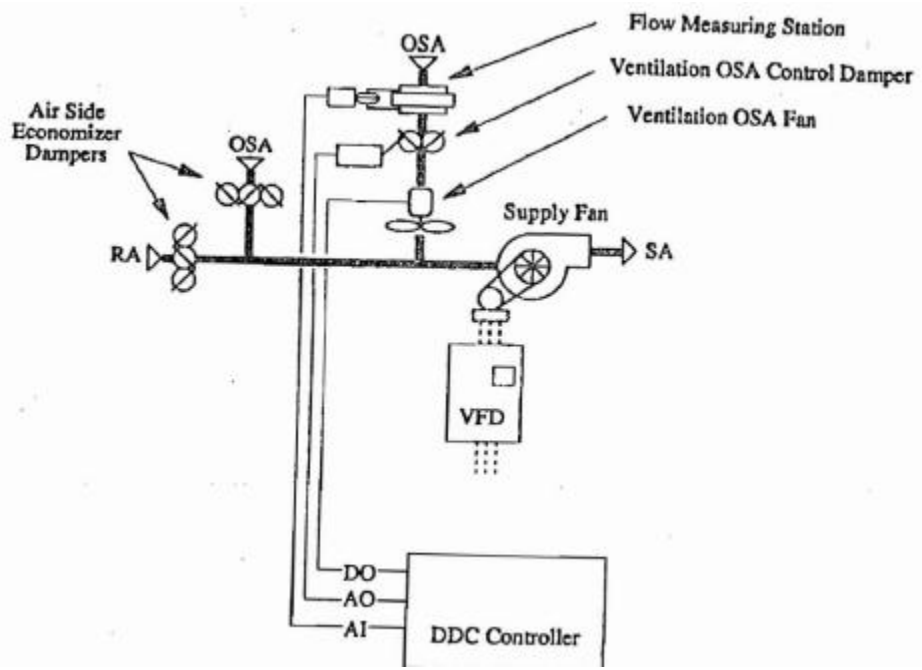
- Local control strategies
  - Minimum outside air control
  - ~~Mixed air control~~
  - Static pressure control
  - Variable air volume terminal control
  - Supply air temperature reset control
  - Enthalpy control
- For central plant systems
  - Hot water reset control
  - Chilled water temperature reset
  - Coolin tower fan speed control

Direct Digital control Application strategies

- Variable pumping strategies for chilled water system
  - Single chiller configuration
  - Multiple chiller configuration
- Direct digital control of Air handling system
  - Variable air volume systems
  - ~~Dual duct system~~
  - ~~Multizone system~~
  - ~~Package unit systems~~

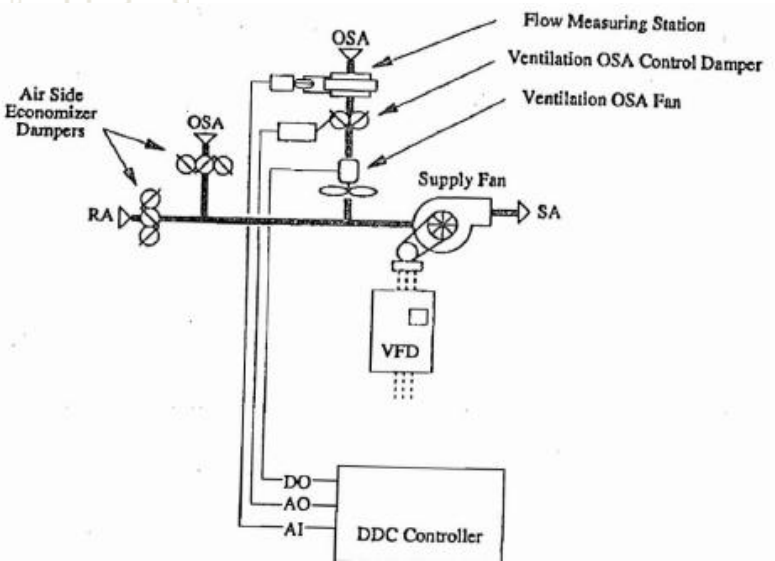
Minimum Outside air control

- To take in OA when it is at required temperature
- Two types
  - Pressure dependent
    - Intake volume changes with fan speed
  - Pressure independent
    - Control sequences that gurantee a minimum volume of ventialtion regardless of the fan speed



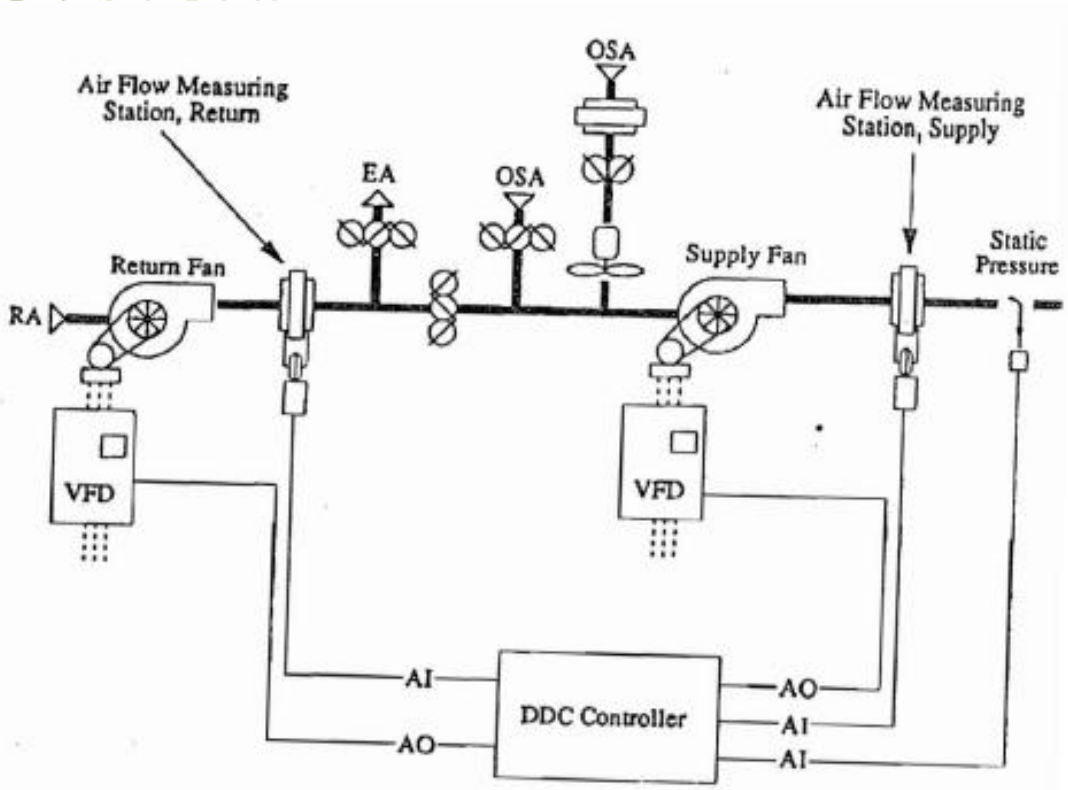
Minimum Outside air control

Point Description	(DO)					(AO)					(DI)				(AI)						
	Start/Stop	Open/Close	Other	Alarm		Position Control	Speed Control	Control Point Adjust	Other	Alarm	Status	Other	Alarm		Temperature	Pressure	Flow	Speed	Volume	Other	Alarm
Outside Air Volume (CFM)																			X		
Start/Stop Command, OSA Fan	X																				
Position Control, OSA Damper						X															



Static pressure control

- Purpose of static pressure control is to maintain minimum amount of static pressure in duct system to support proper air distribution



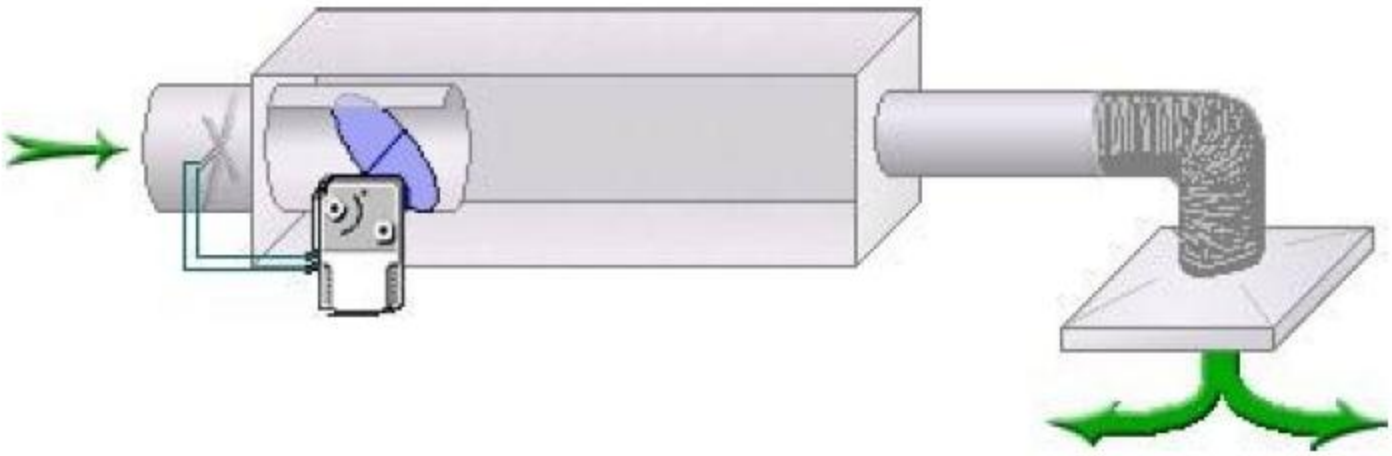
Two primary modes of controls are

- Duct static pressure
- Building static pressure

Supply and return air flow control

Variable air volume control

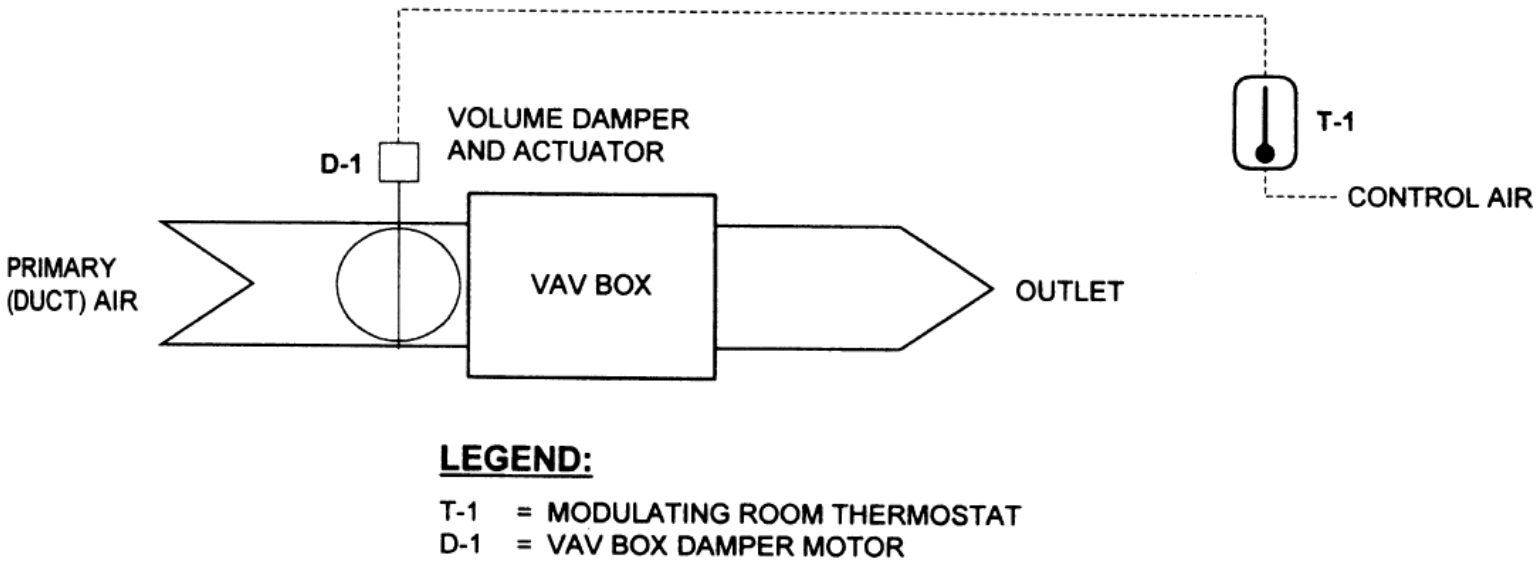
- The purpose of VAV terminal control is to modulate the cooling supply air to meet the minimum cooling requirement of the space





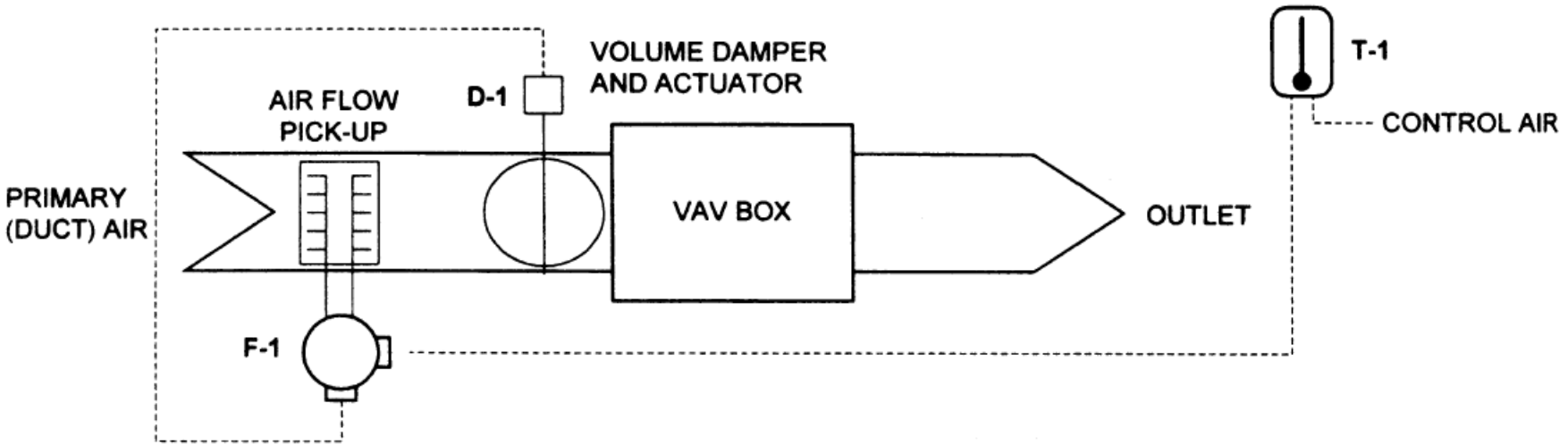
Variable air volume control-cont'd

- VAV box which is pressure dependant(no pressure control)



Variable air volume control-cont'd

- VAV box which is pressure independant(with pressure control). Flow controller controls the minimum flow requirement

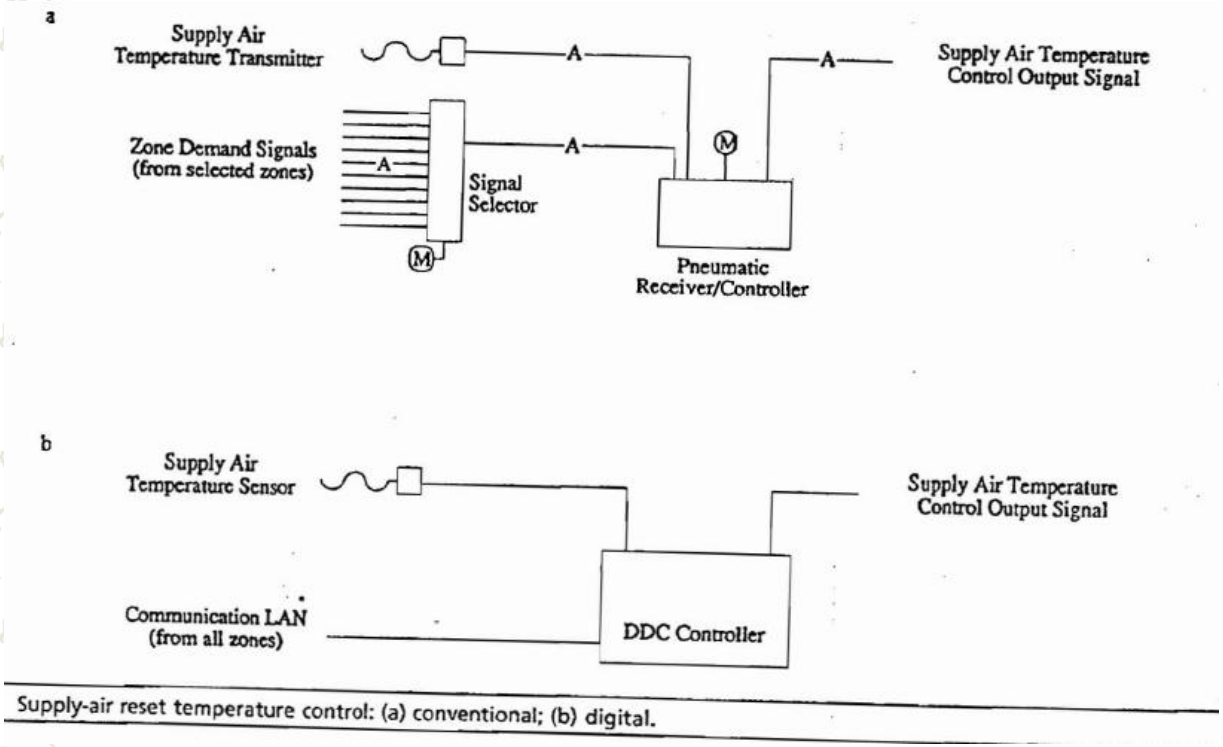


LEGEND:

T-1 = MODULATING ROOM THERMOSTAT
D-1 = VAV BOX DAMPER MOTOR
F-1 = FLOW CONTROLLER

Supply air temperature reset control

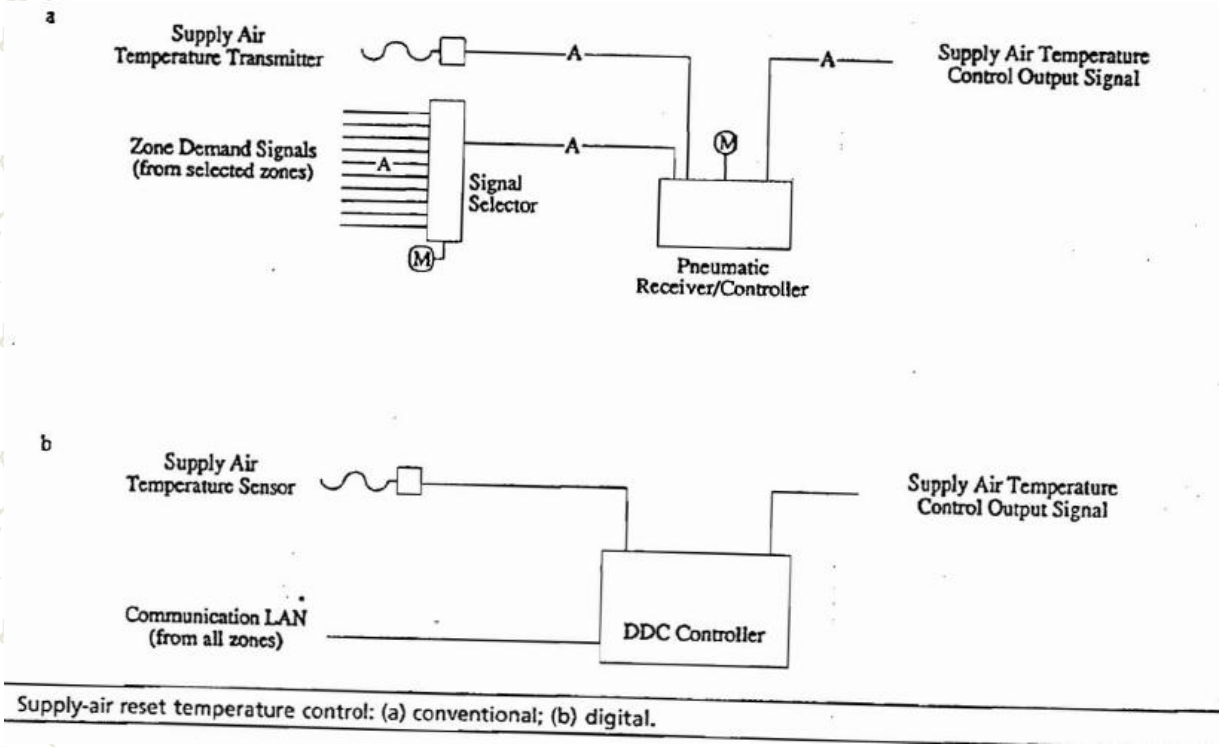
- The purpose of supply air reset temperature control is to reset the supply air temperature when there is a reduced demand for cooling.



Supply-air reset temperature control: (a) conventional; (b) digital.

Supply air temperature reset control

- The purpose of supply air reset temperature control is to reset the supply air temperature when there is a reduced demand for cooling.



Supply-air reset temperature control: (a) conventional; (b) digital.

Enthalpy control

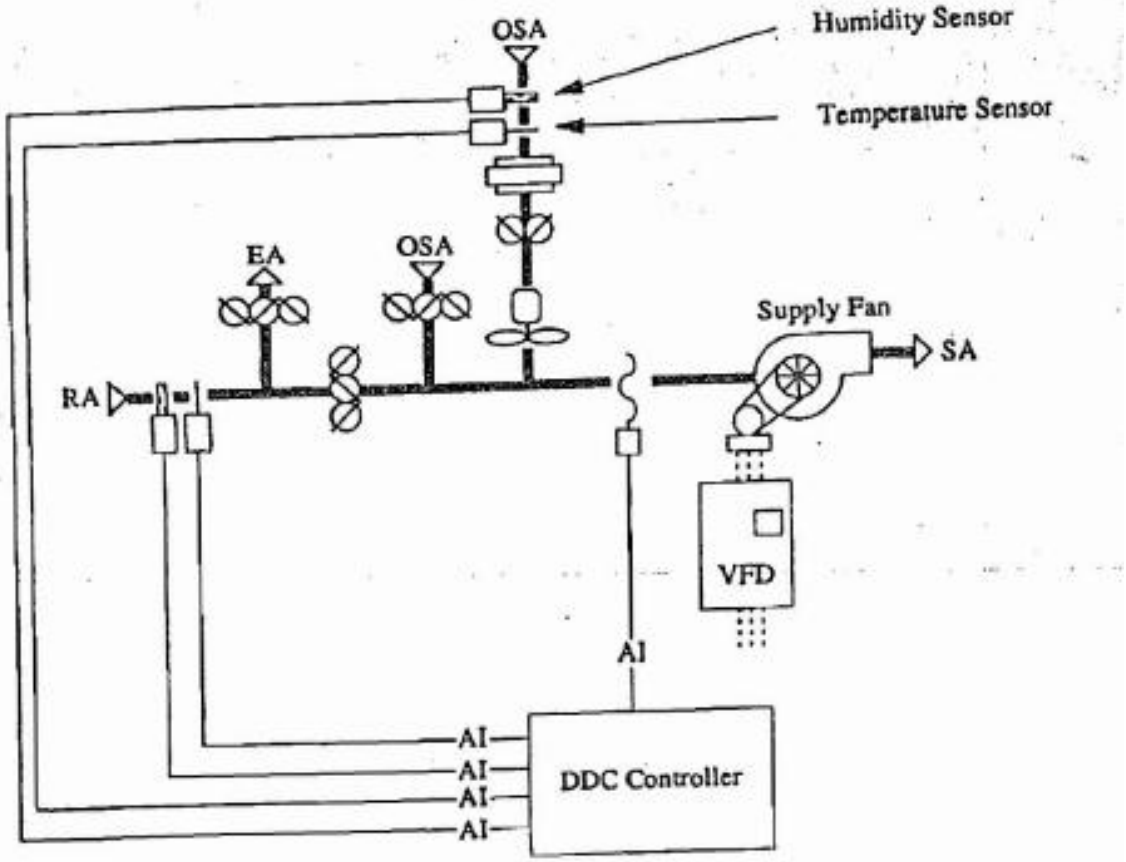


Figure 6-6a Components for enthalpy control of economizer dampers.

Hot-water reset control

- Adjustment of hot water temp based on the demand for heat. Demand for heat varies in inverse proportion.

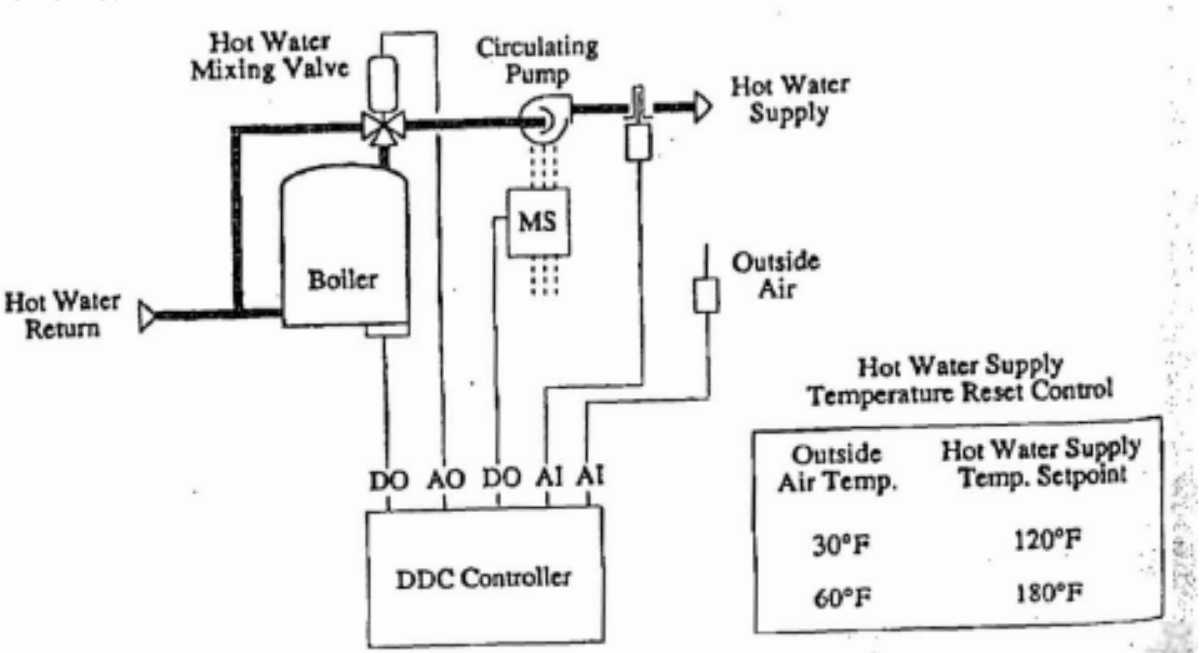


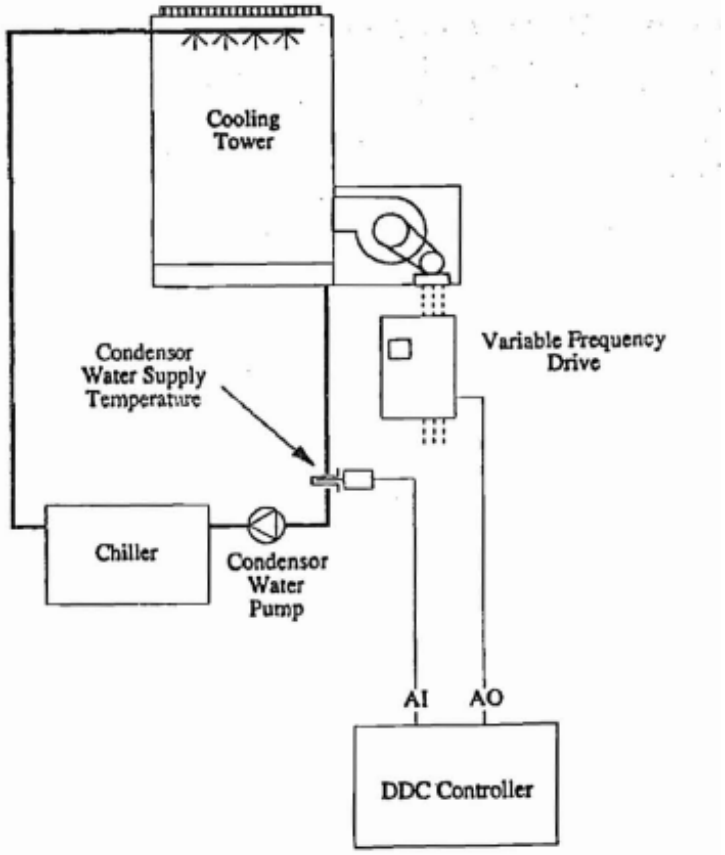
Figure 6-7a Heating hot-water reset temperature control.

Chilled water temp reset control

Temperature reset of the chilled-water supply (CHWS) temperature can lower the operating cost of chiller plants. An aggressive, yet effective, energy conservation strategy using DDC is to reset the CHWS temperature based on zone valve position. The concept of this strategy is similar to the duct-pressure reset strategy discussed earlier. When the position of each chilled-water coil valve is known, the valve operating under the greatest cooling demand can reset the CHWS temperature upward until this valve reaches 100% of its stroke. As the water temperature gradually rises, the other valves in the system modulate open to satisfy their zone loads, and should operate more efficiently because they are operating near the top end of their stroke. If this strategy is incorporated into the initial design of the chilled-water distribution system, smaller chilled-water pumps and line-sized control valves can be used, yielding a first-cost saving opportunity.

Cooling Tower fan speed control

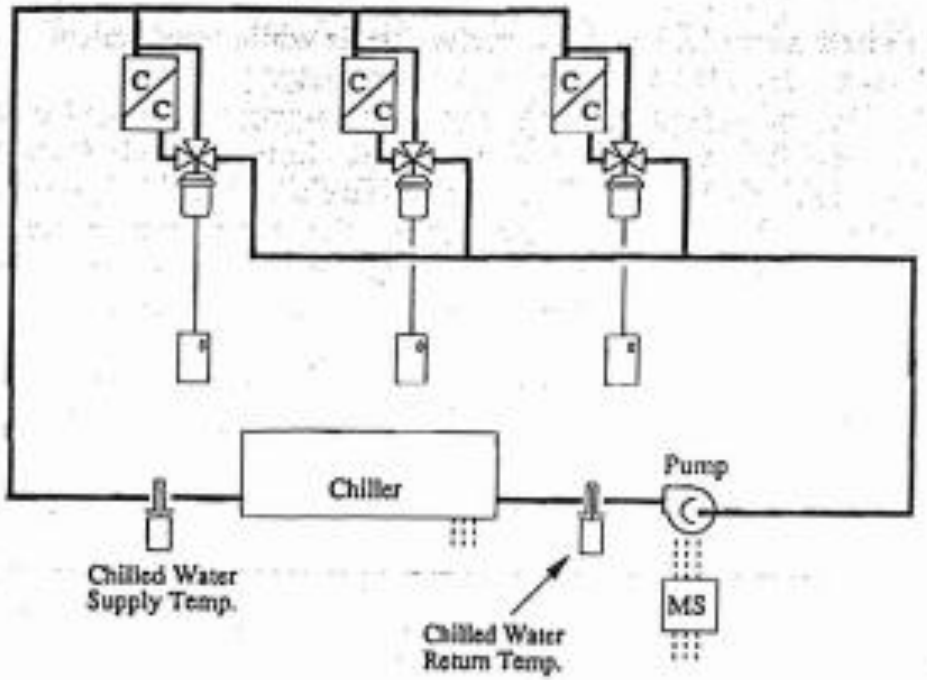
- Adjustment of hot water temp based on the demand for heat. Demand for heat varies in inverse proportion.



Control strategy: When demand for condenser-water temperature is reduced as a result of reduced cooling load, modulate the speed of the tower fan to maintain the condensor water set point

Variable pumping strategies for chilled water system

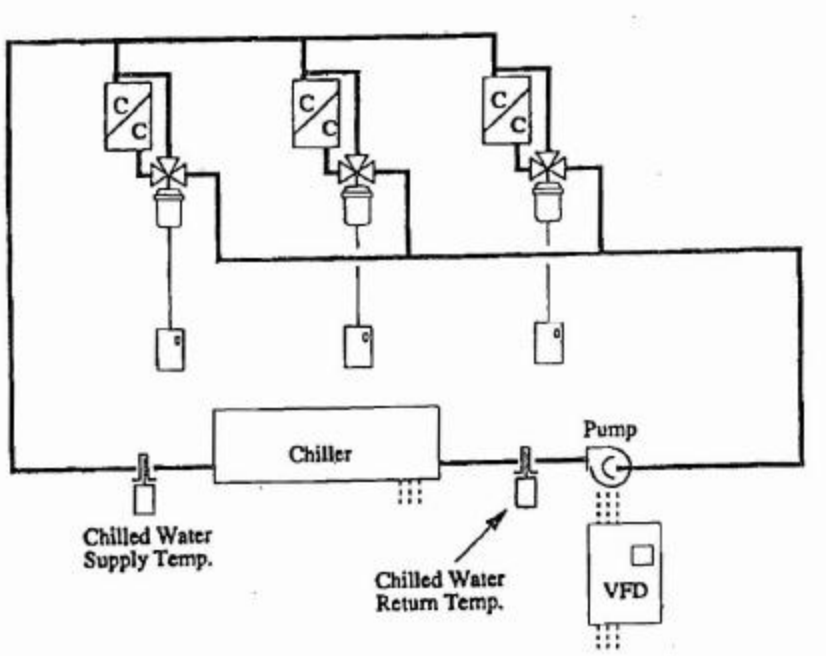
- Single Chiller with three way valves at load.



Most large chilled water system have a diversity of 50% to 60%. This contributes heavily to the energy wasted in recirculating unused water.

Variable pumping strategies for chilled water system

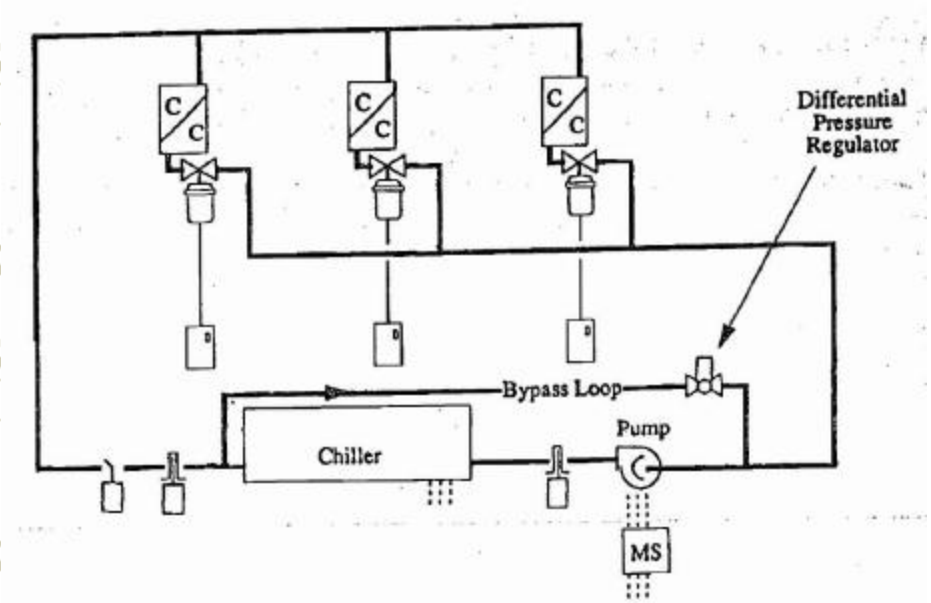
- Single Chiller with three way valves at load and VFD.



Though VFD saves energy by minimizing the unused circulating water, the less water to chiller can create problems.

Variable pumping strategies for chilled water system

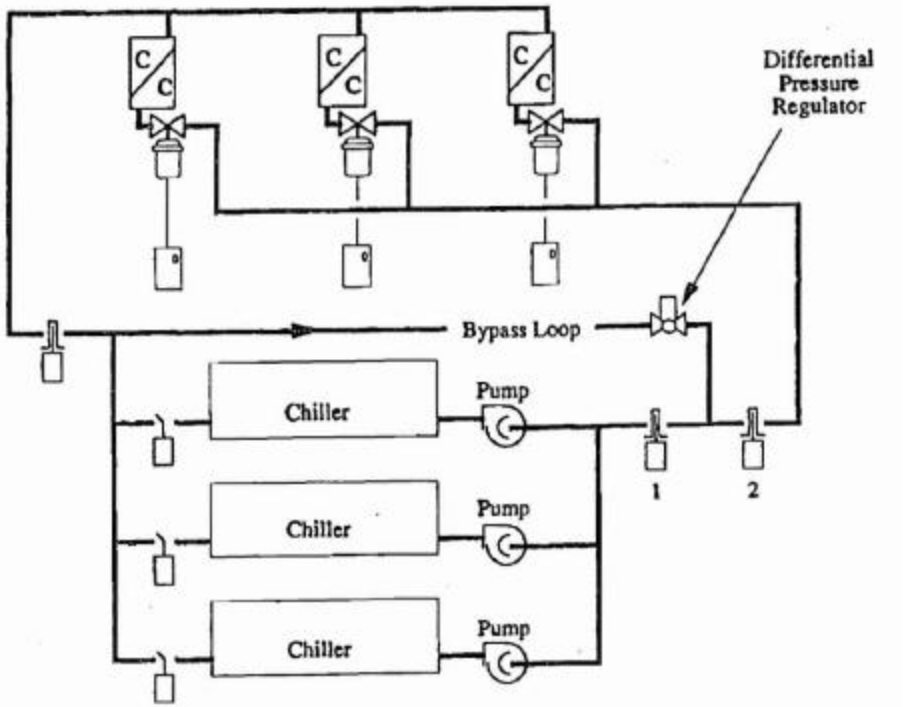
- Single Chiller with two way valves at load and By pass circuit
- Constant chiller flow, variable system flow



Problems of low water flow to chiller is solved
NO VFD, therefore no saving at CHWP
Saving by chiller staging

Variable pumping strategies for chilled water system

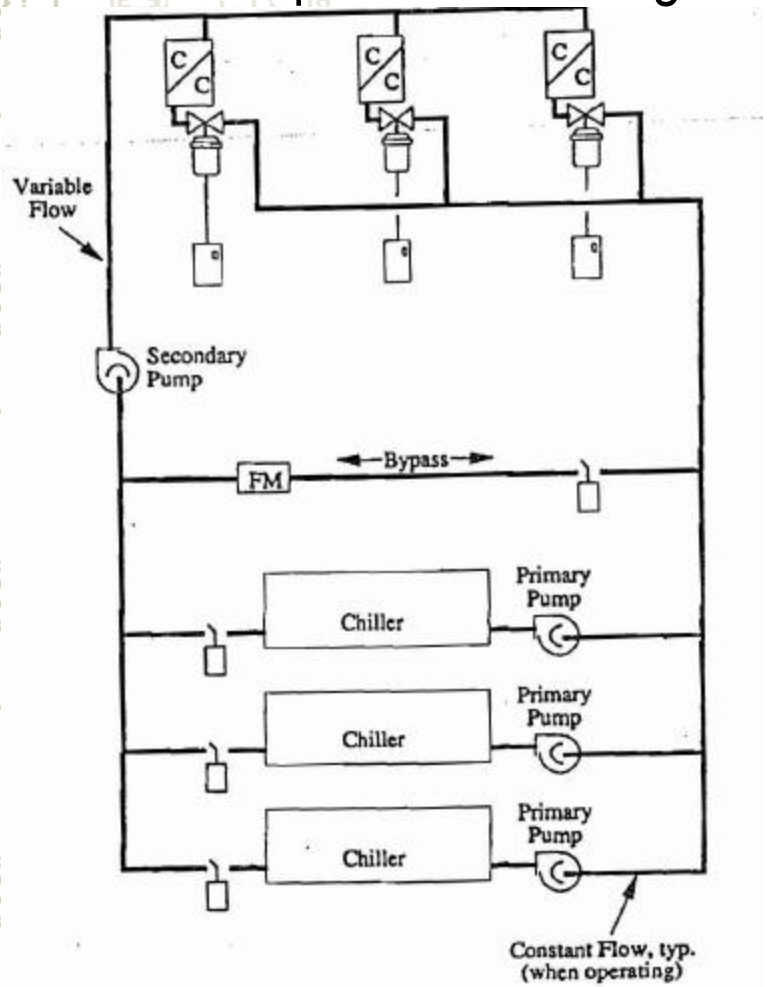
- Multiple Chiller configuration- Constant chiller flow, Varaible system flow



• Addition of a temp sensor
•The relationship between return temp and supply temp indicates a surplus or deficit of water supply compared with water demand.
•Measuring the system load by valve postions.

Variable pumping strategies for chilled water system

- Multiple Chiller configuration-Primary and secondary chilled water supply loops

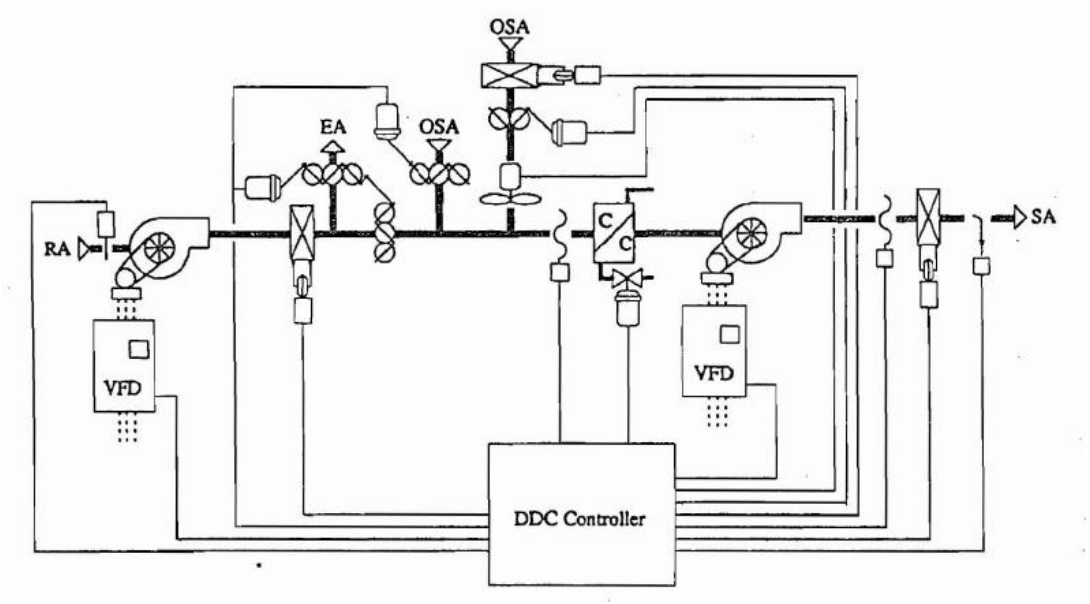


- ❑ Chiller may operate independently under constant flow conditions
- ❑Distribution system will operate for the building demand with variable flow
- ❑Flow switch to indicate the direction of flow
- ❑Flow meter measure the flow

- •Chiller plant operate under highly efficient constant flow variable temp
- •Building load to operate in a varaible flow constant temp which is most efficient in pumping capacity

Direct Digital control of air handling system

- Variable air volume system



- Minimum outside air control, supply reset control, duct static pressure control, and zone terminal control all act in concert to provide right quantity and temeperature of air to the occupied space.

Monitoring strategies for Building Managment

- DDC system have databases containing valuable information on the operating characteristics of the system they control.
- This information is useful in improving efficiency and the effectiveness of a DDC system

- Power consumption monitoring
  - Trend Logs
  - Operation reports
  - Maintanance time reminders
  - Energy Management reports
- Supervisory control strategies
  - Time scheduled control
  - Optimal start/stop
  - Duty cycle control
  - Load shedding
  - Temperature setback
  - Optimization control routines

Power consumption Monitoring

- Trend Logs
  - Is a sequential listing of the current values of points on the control system.
  - Good for verifying the proper operation and for troubleshooting

Power consumption Monitoring

- Operation Reports
  - Is like a trend log except that it provides information on a single system loop
- Maintenance time reminders
  - To organize, monitor and acknowledge the activities associated with a comprehensive maintenance program
    - Equipment operation status
    - Total equipment runtime
    - Date of last service

Power consumption Monitoring

- Energy Management reports
  - Historical data on electrical usage – to evaluate the efficiencies of the systems
  - Energy use profiles can be developed for each system
  - An important usage of energy management reports is for cost justification for EMS investments.

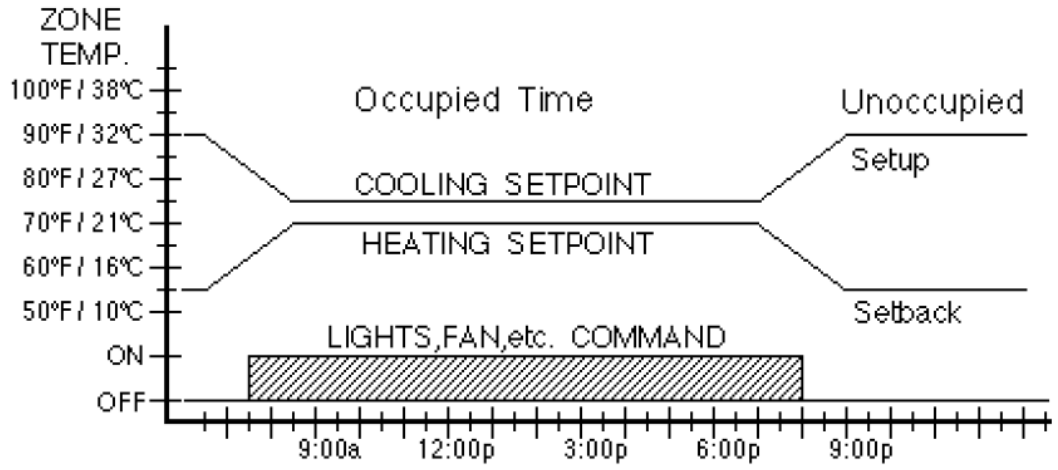
Supervisory Control Strategies

- Monitoring of activities of local control loop and overriding local control
- It has capability to override local programmed control routines when necessary to achieve overall objective of the control system

Time scheduled control

- Operation of systems based on the building occupancy
  - Reduced inrush current

Time-Programmed Commands

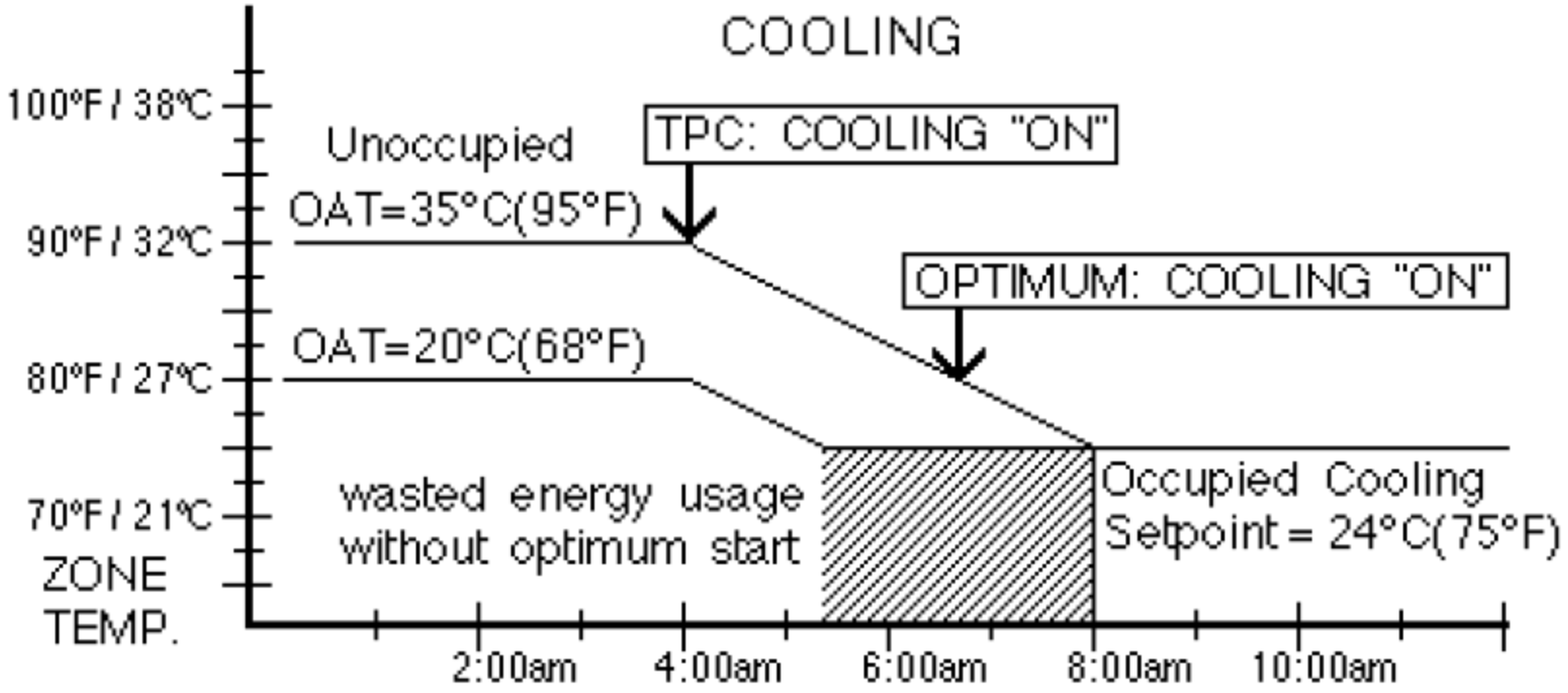


Optimal Start/Stop

- Optimal start/stop strategies are software – driven routines that calculates optimal time of day and operating conditions for initiating the system operation
  - Morning warm-up by referencing historical morning temperatures over a defined period of time
    - Not starting too early-Wastes energy
    - Not starting too late - Discomfort

Optimal Start/Stop

Optimum Start / Stop

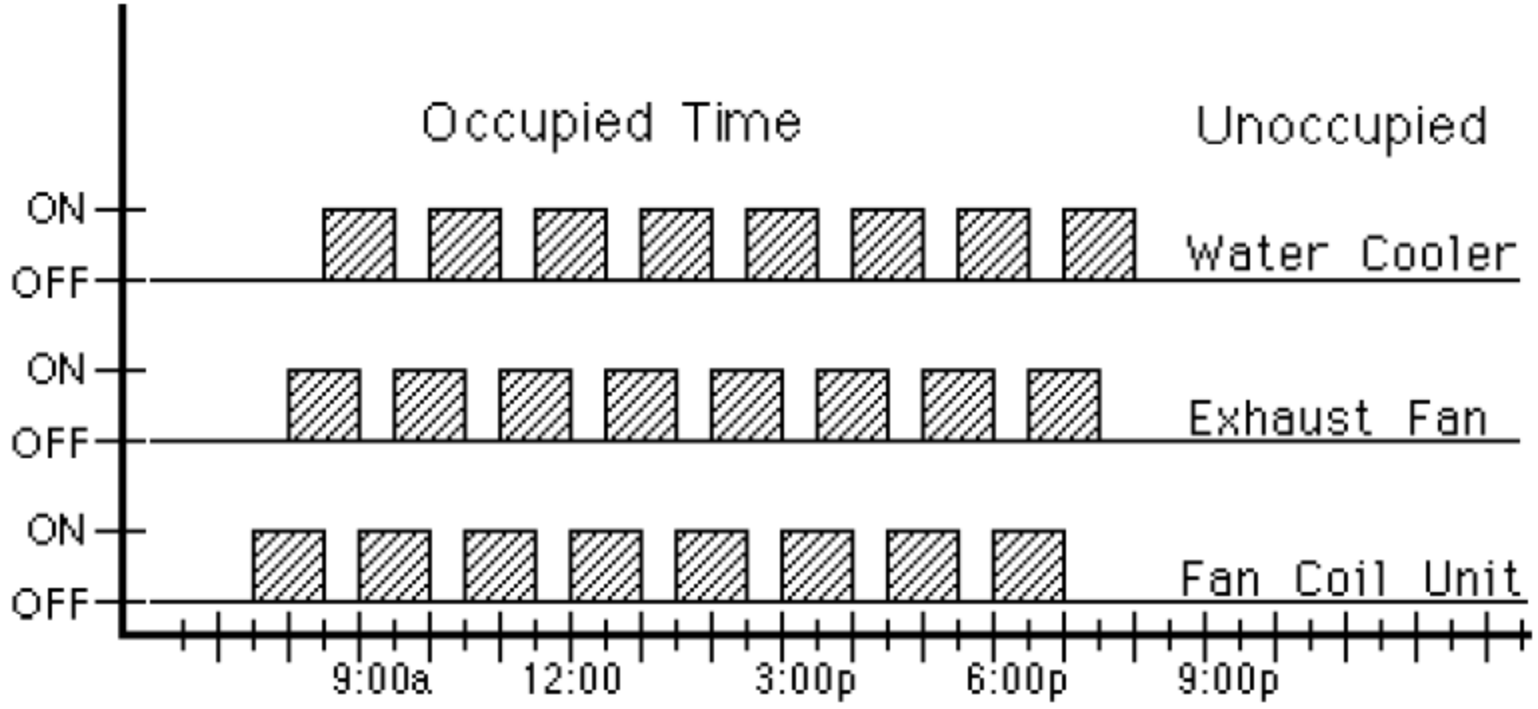


Duty Cycle Control

- Duty cycling is a strategy that involves shutting down selected HVAC systems for a fraction of their normal run time to reduce the overall electricity demand.
- Negative aspect of it, wear and tear caused by excessive on/off starting
- It is much wiser to reduce the speed rather than turning them off.

Duty Cycle Control

Duty Cycling

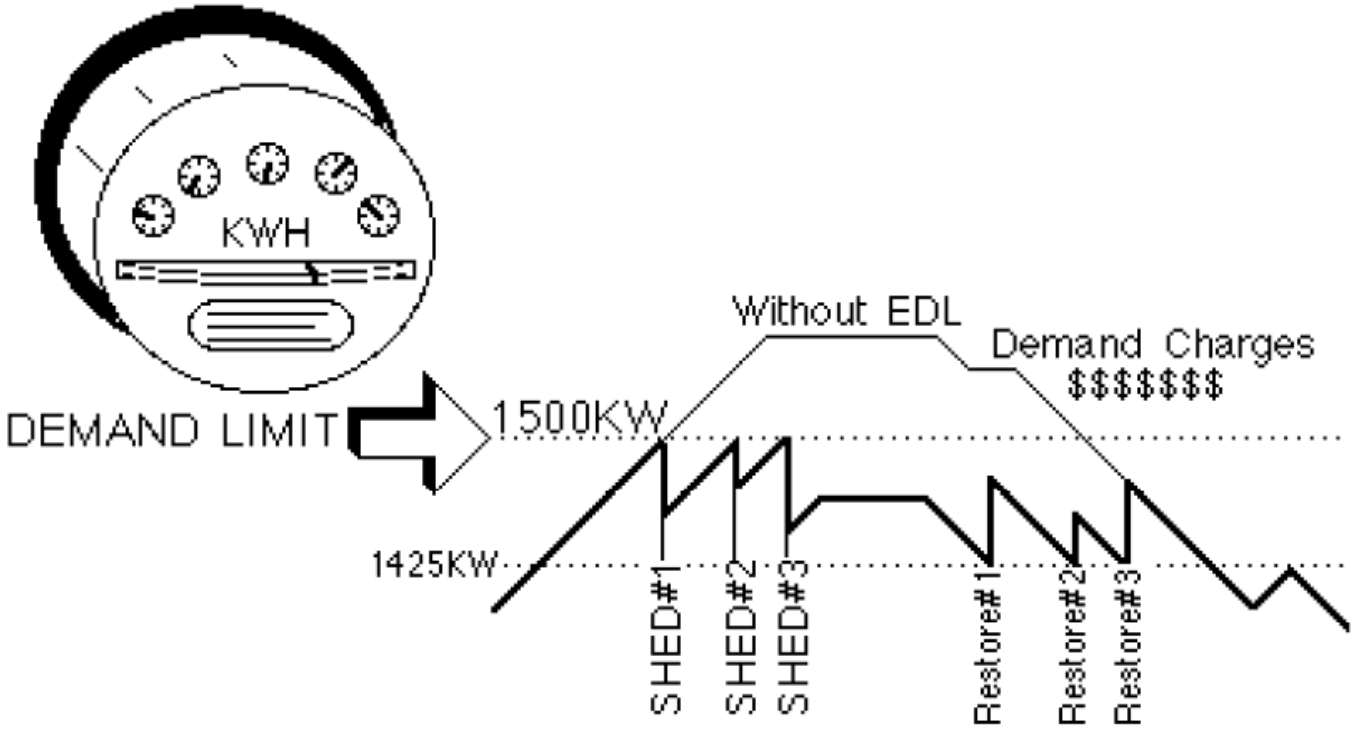


Load Shedding

- Load shedding is a strategy that is similar to duty cycling control except that the command to start or stop a piece of mechanical equipment is a function of the overall building electricity demand.

Load Shedding

Power Demand Limiting is also referred to as EDL.



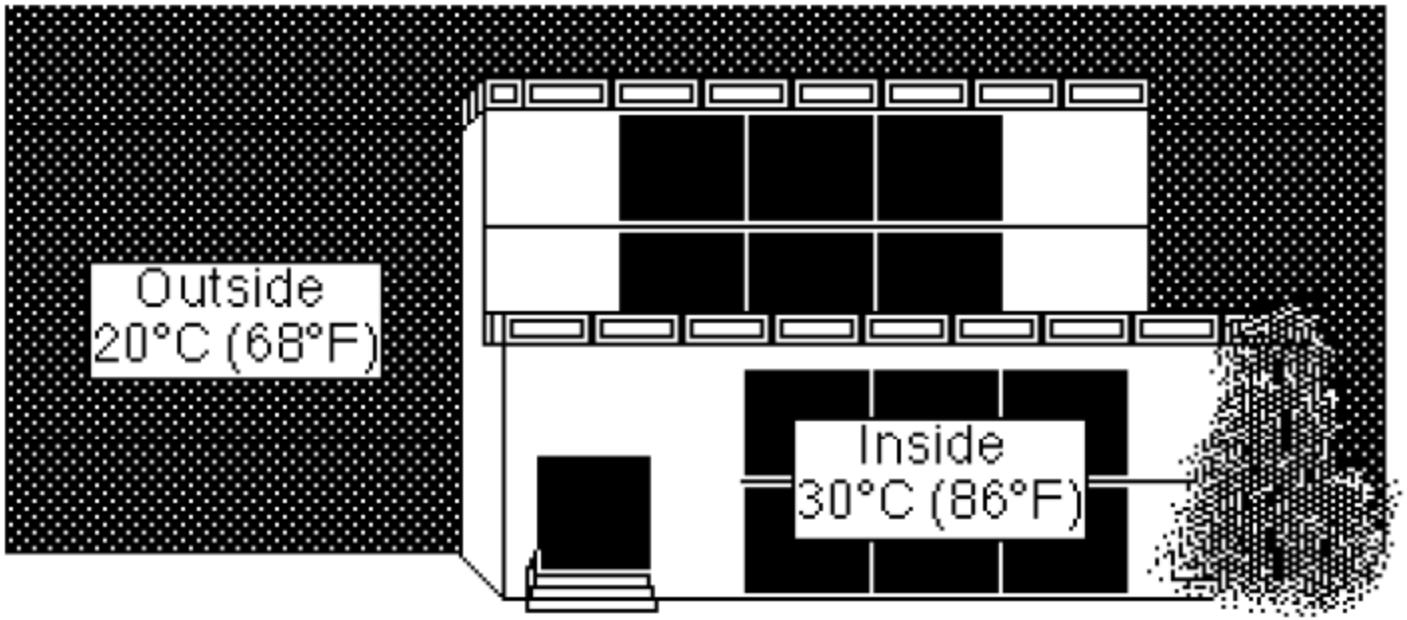
ELECTRIC DEMAND LIMITING (EDL),

Temeprature Setback

- Also called night setback or night depression
  - In colder climates
    - Is used to reduce the temperature of the building during unoccupied time – setting at lower set point, usually 10 to 15 deg lower
    - Can easily bring up the temp in the morning
  - In warmer climates
    - use cool evening air to pre cool the building

Temperature Setback

Night Purge Cycle



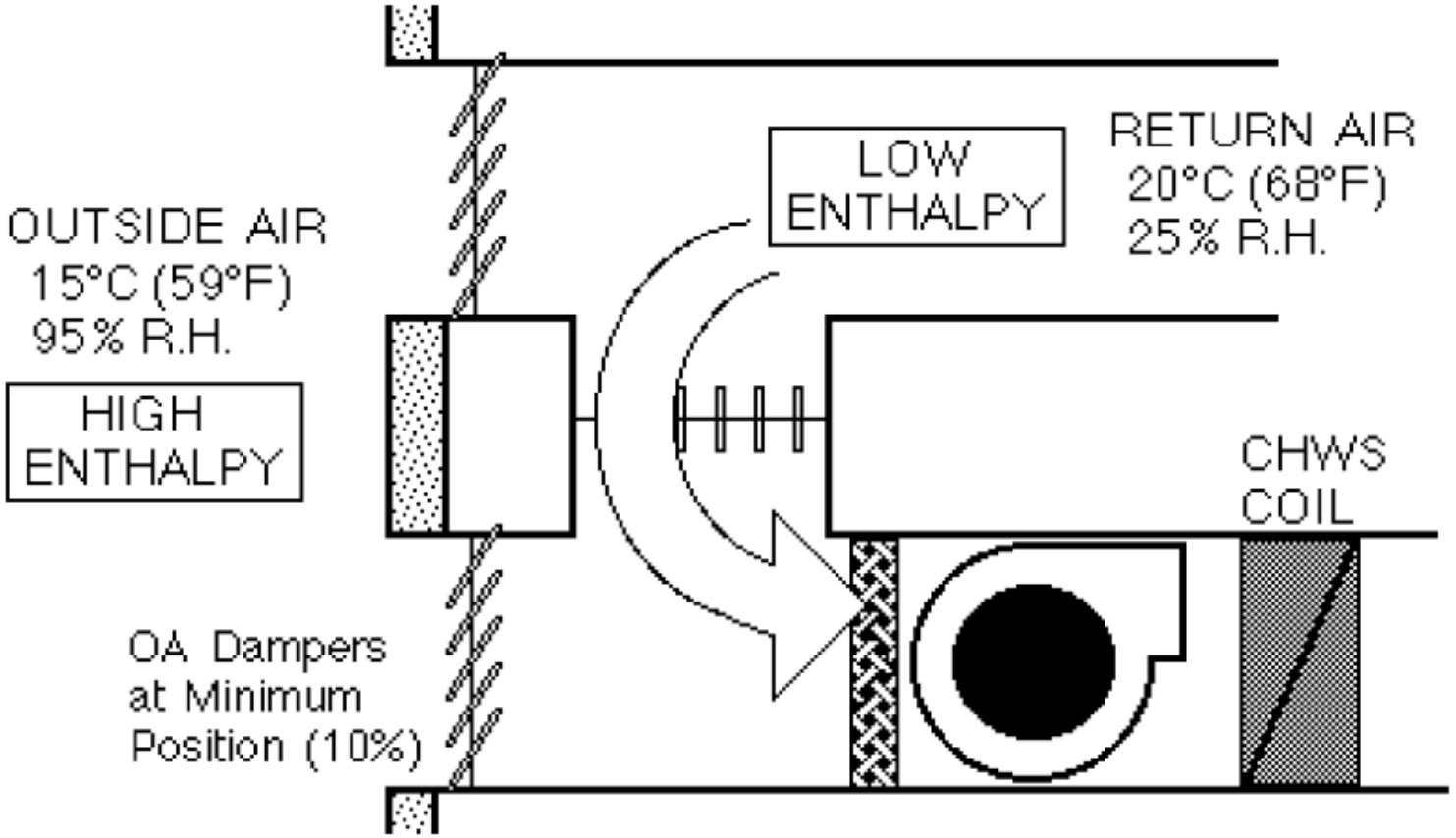
Optimization control routines

- Optimize and improve the performance
  - Improving the efficiency of the equipment being controlled
  - Improving the system efficiency by matching the device to its load(Set point change, etc)

  Specially for Large equipment such as Chillers and Boilers

Optimization control routines

Enthalpy Optimization



System Design Methodology

Control of an HVAC System

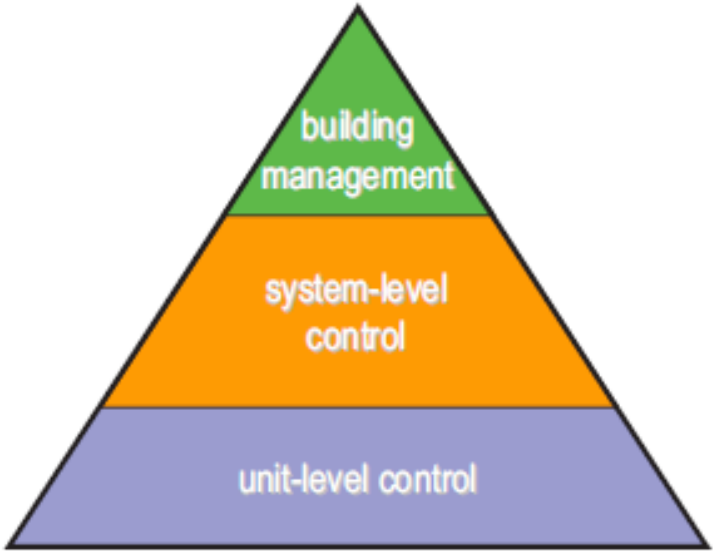


Figure 27

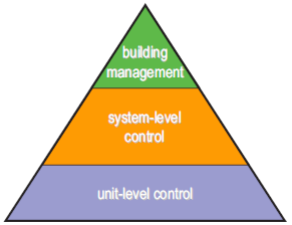
System Design Methodology

Benefits of Unit-Level Control

- Stand-alone control
- Safeties, alarms, and diagnostics
- Installed, tested, and commissioned in factory

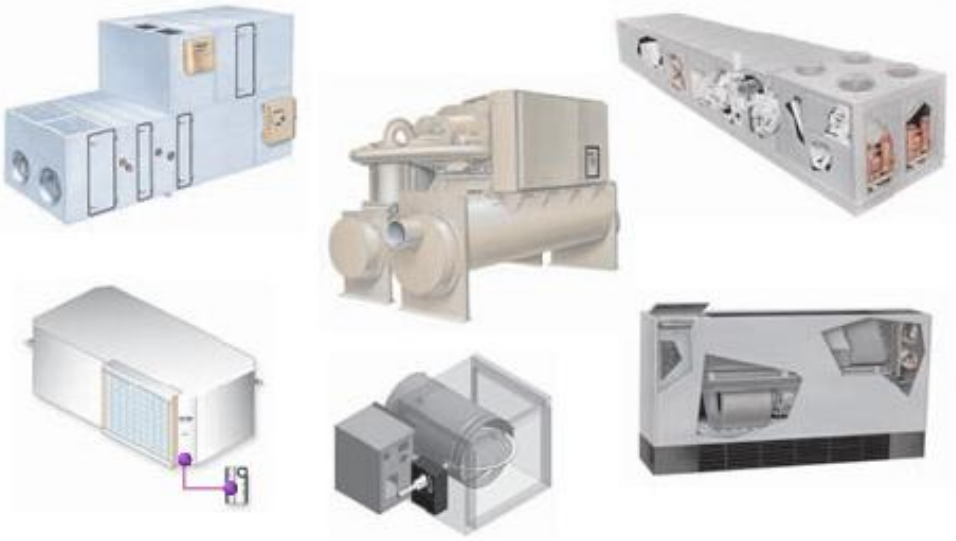


Control of an HVAC System

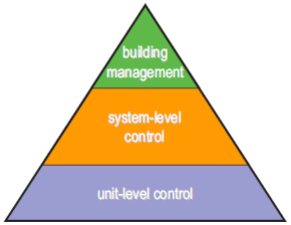


System Design Methodology

Unit-Level Control



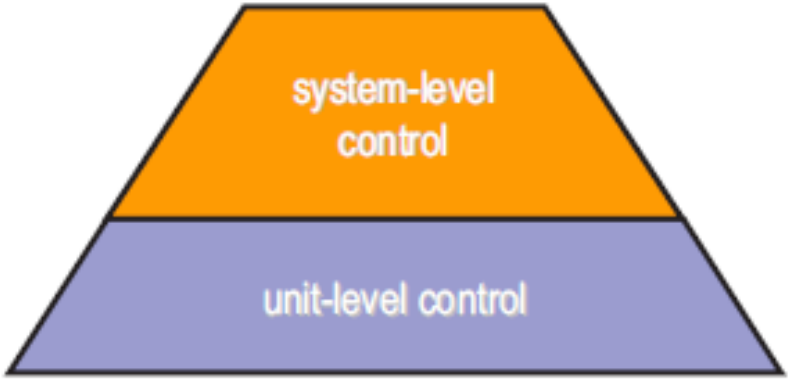
Control of an HVAC System



- Primary function is to protect the equipment
- Standalone-capable of performing even if no support from system level

System Design Methodology

System-Level Control



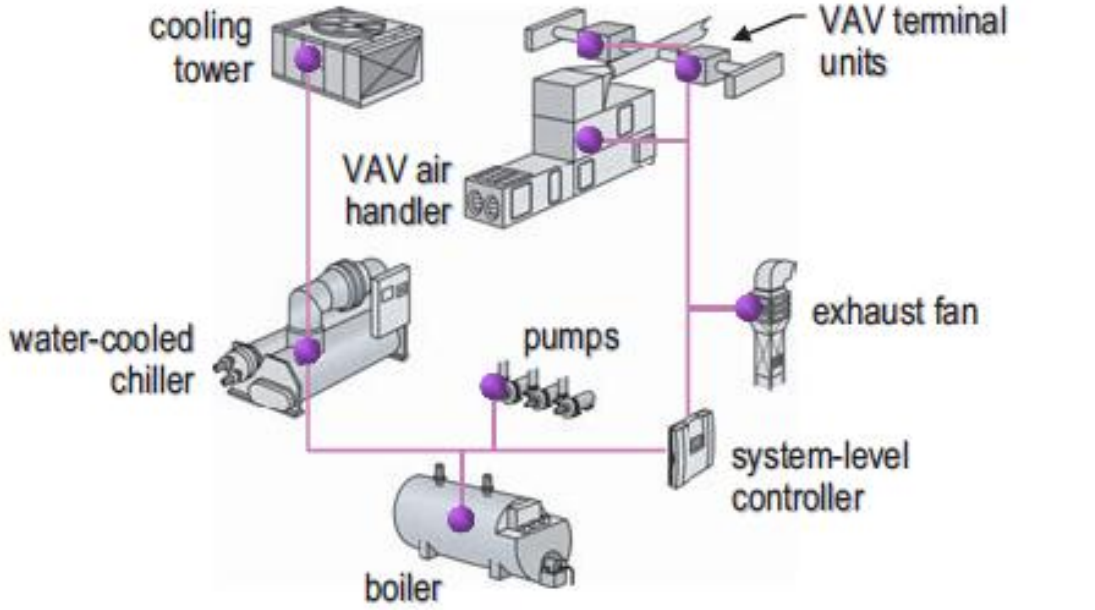
Control of an HVAC System



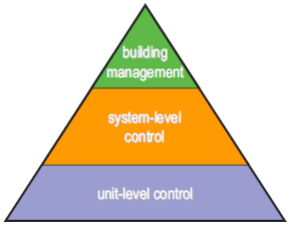
•Provides intelligent , cordinated control

System Design Methodology

Chilled-Water VAV System



Control of an HVAC System



System Design Methodology

Building Management



Control of an HVAC System



System Design Methodology

Responding to Comfort Complaints

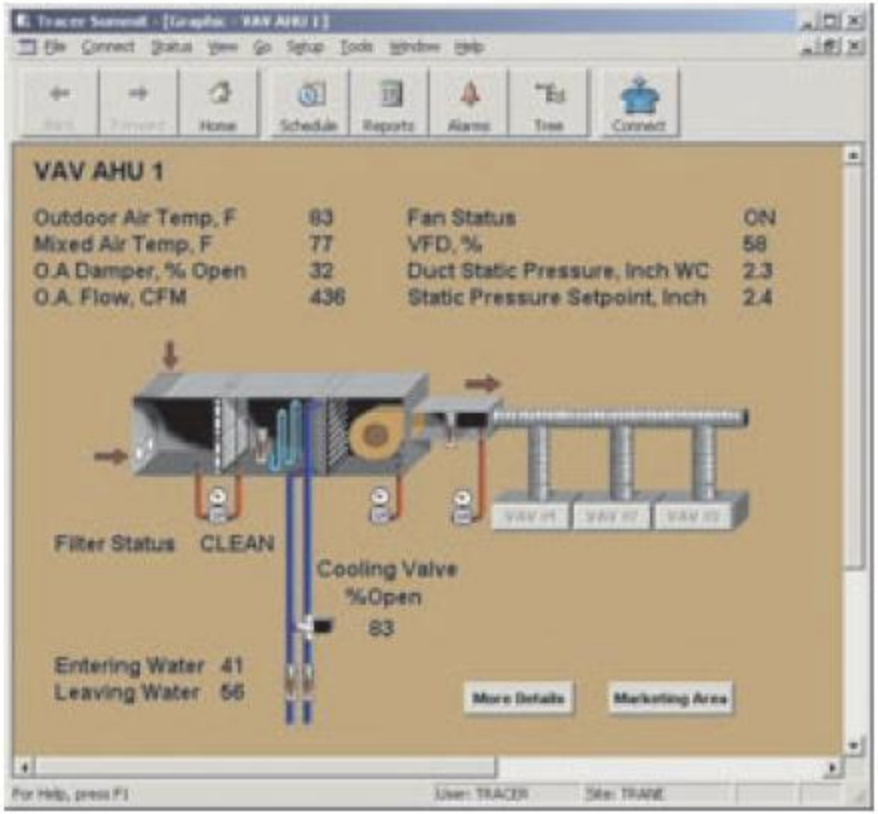


Control of an HVAC System

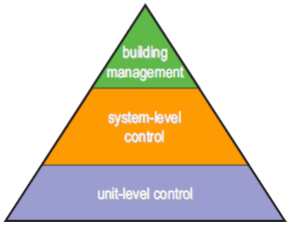


System Design Methodology

Graphical User Interface

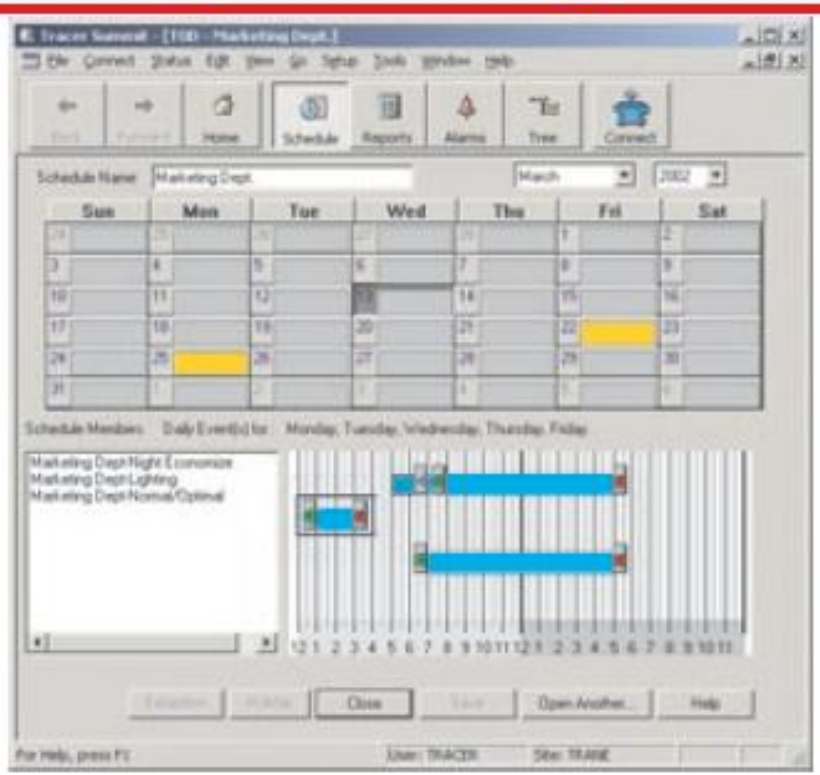


Control of an HVAC System

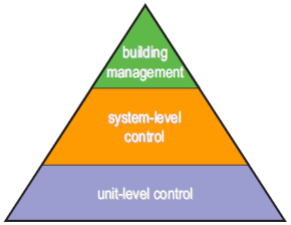


System Design Methodology

Time-of-Day Scheduling



Control of an HVAC System

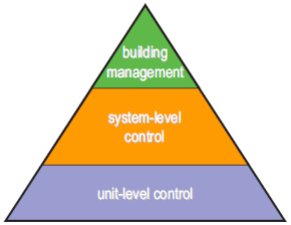


System Design Methodology

Centralized Alarms and Diagnostics



Control of an HVAC System



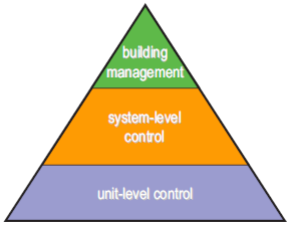
System Design Methodology

Remote Access

- Alarm notification
- Dial-in access



Control of an HVAC System



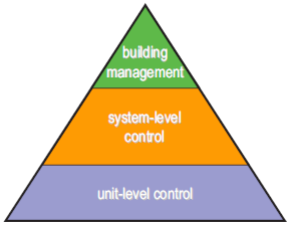
System Design Methodology

Reports

- Trend logs
- Alarm activity
- Building energy use
- After-hours tenant occupancy
- Equipment diagnostics and performance
- Commissioning



Control of an HVAC System



System Design Methodology

Preventive Maintenance



Control of an HVAC System



System Design Methodology

Connecting Multiple Sites

- Centralized monitoring
- Diagnose or predict problems with systems
- Combine energy usage for negotiating energy contracts



Control of an HVAC System



System Design Methodology

- The design of conventional control systems
  - Create HVAC schematic
  - Identify sensors and transmitters
  - Identify controllers
  - Identify controlled devices
  - Identify the appropriate interconnecting media
  - Specify sequences of control operation

System Design Methodology

- The design of DDC systems
  - Create the HVAC system schematic
  - Identify each control points
  - Identify sensors and transmitters
  - Identify controlled devices
  - Identify interconnecting media
  - Determine the sequence of control operation
  - Identify the field computer hardware
  - Identify the host computer hardware

System Design Methodology

- The simplest way to organize the devices, sequences, and hardware requirements of a large control system is to use the point list.

System Design Methodology

Point Description	(DO)					(AO)						(DI)				(AI)							
	Start/Stop	Open/Close	Other	Alarm		Position Control	Speed Control	Control Point Adjust	Other	Alarm		Status	Other	Alarm		Temperature	Pressure	Flow	Speed	Volume	Other	Alarm

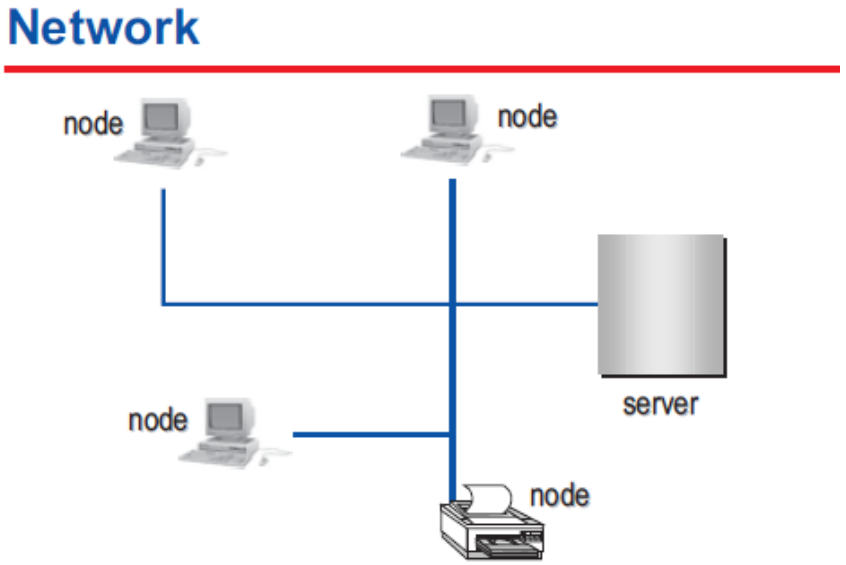
DDC system input/output checksheet.

Interoperability

- Todays trend in BMS is interoperability
- Is defined as the digital communication between independently manufactured product that are designed to the same open, standard communication protocol. It means single controller can monitor and control independently designed, microprocessor based control products.

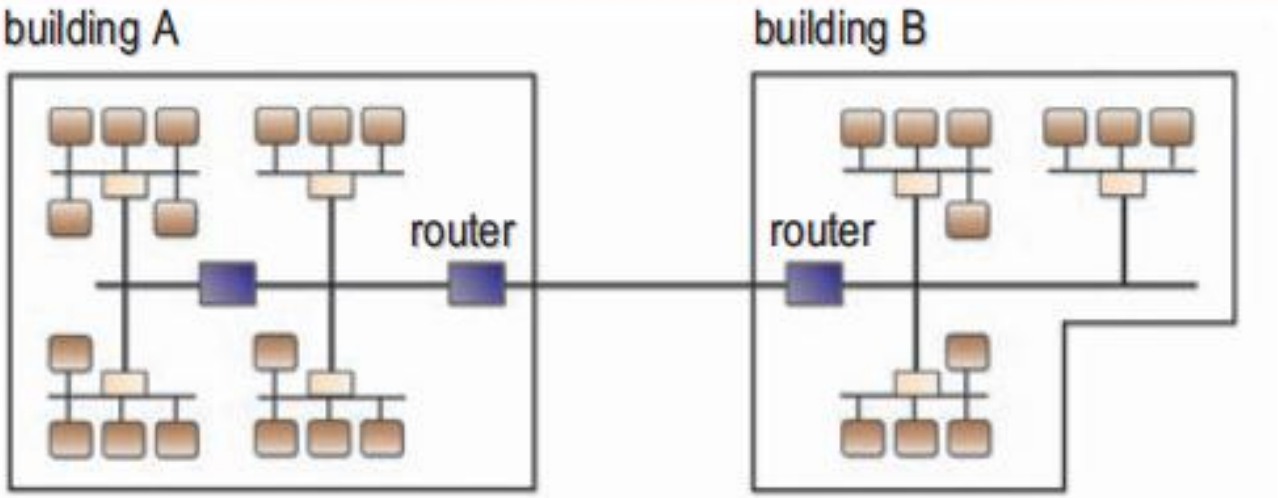
Interoperability

- Network
  - Two or more communication devices are connected so they exchange data
  - Server shares resources
  - Dedicated and shared network



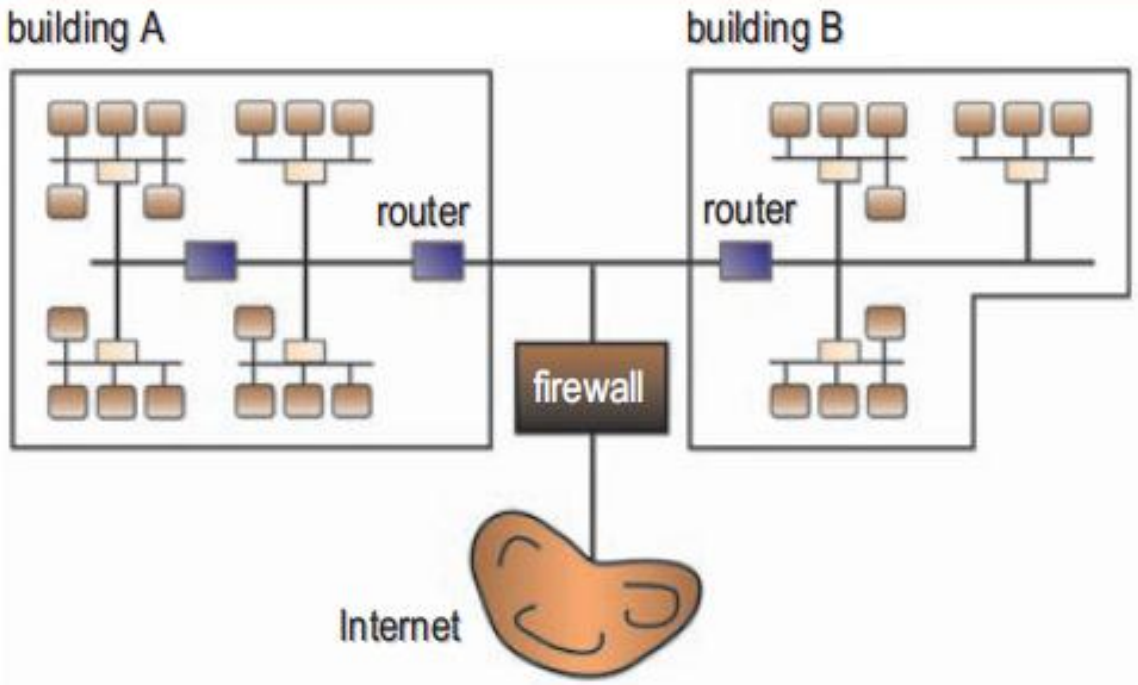
Interoperability

LAN Versus WAN

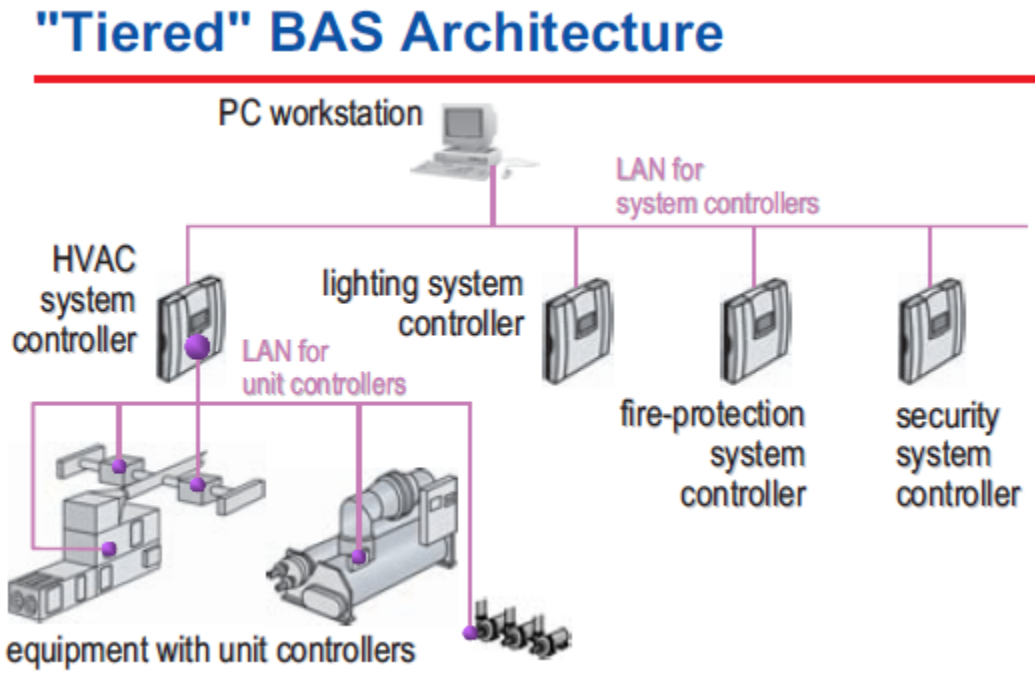


Interoperability

Intranet Versus Internet



Interoperability



- This is the Conventional approch to BAS
- System level controllers are independent
- Reduces the number of interfaces points between subsystems

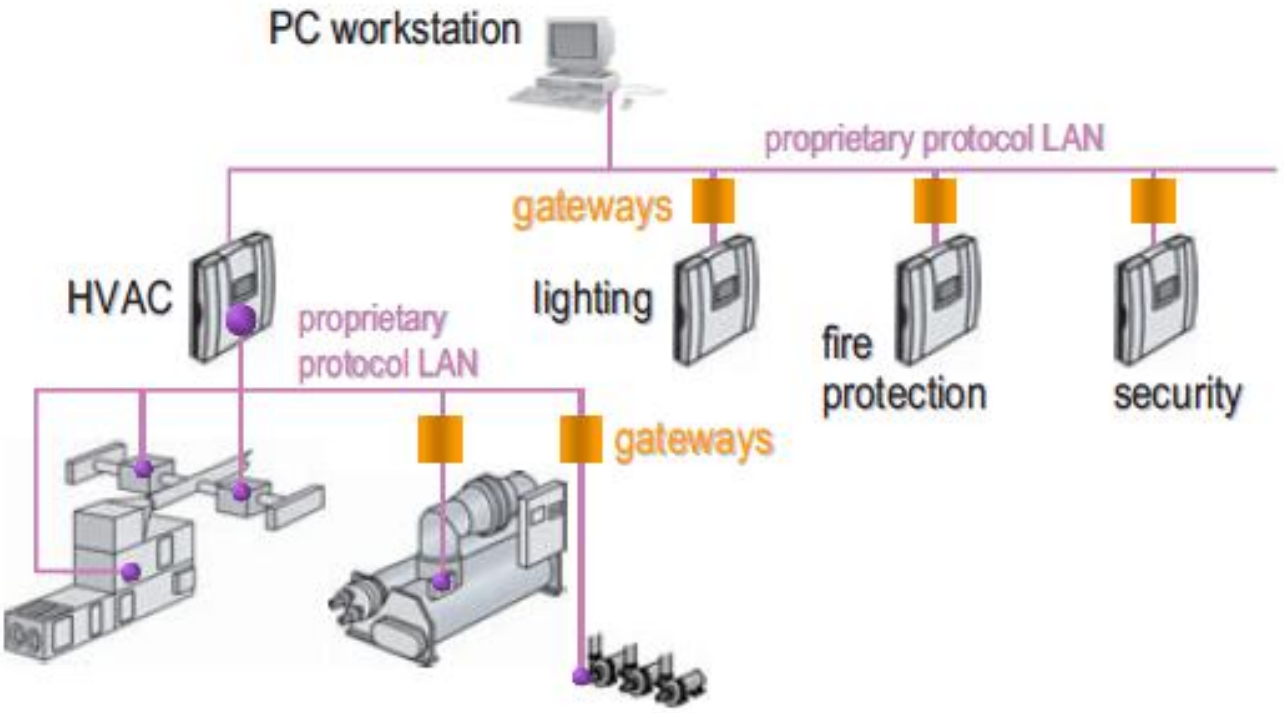
Interoperability

Communications Protocols

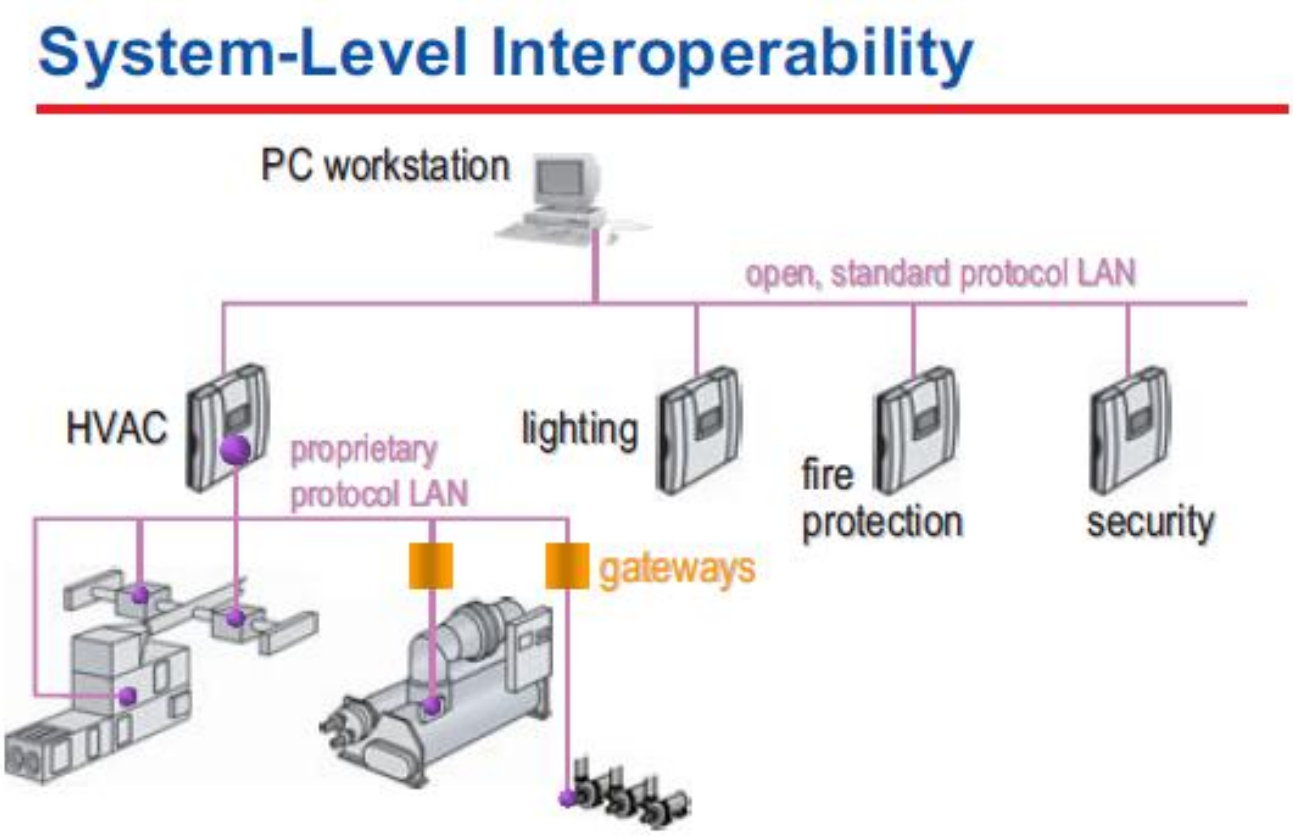
- **Proprietary**
  - Used, produced, or marketed under exclusive legal right of an individual or organization
- **Open**
  - Available to public domain and is shared among vendors
- **Standard**
  - Open protocol that has been formalized by a governing body

Interoperability

Gateways

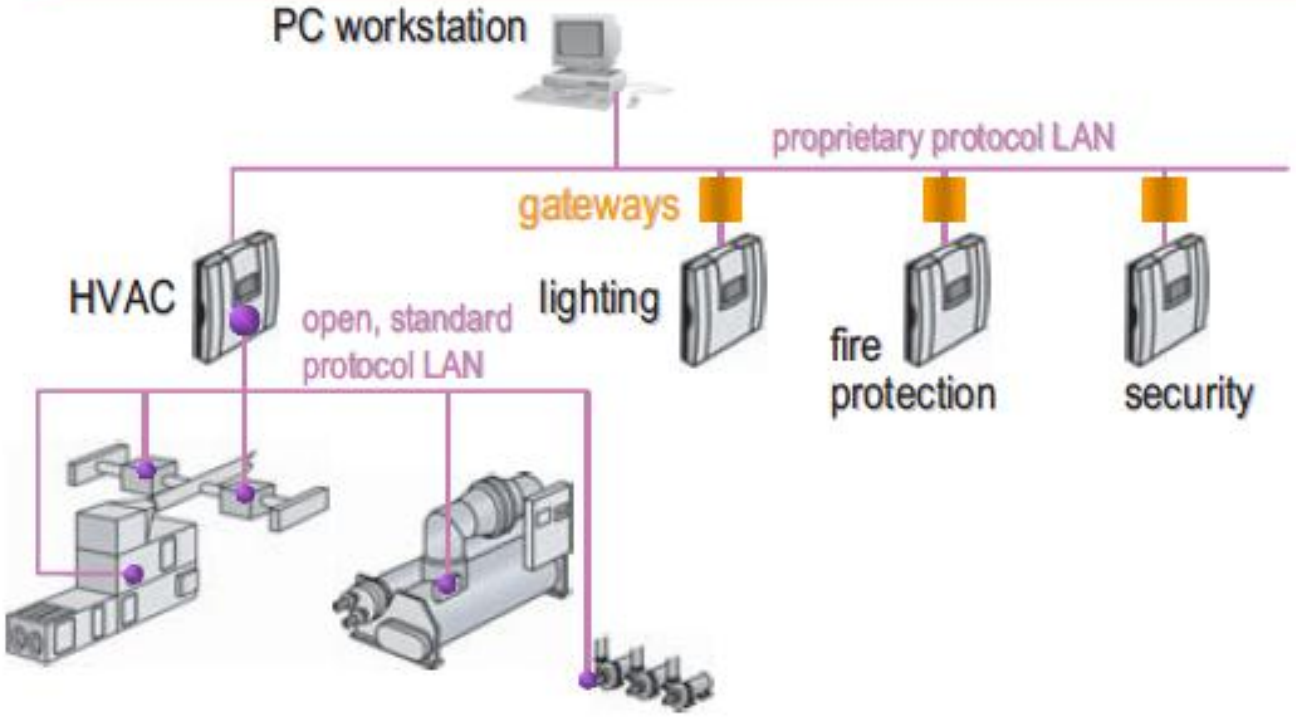


Interoperability



Interoperability

Unit-Level Interoperability



Interoperability

Interchangeability

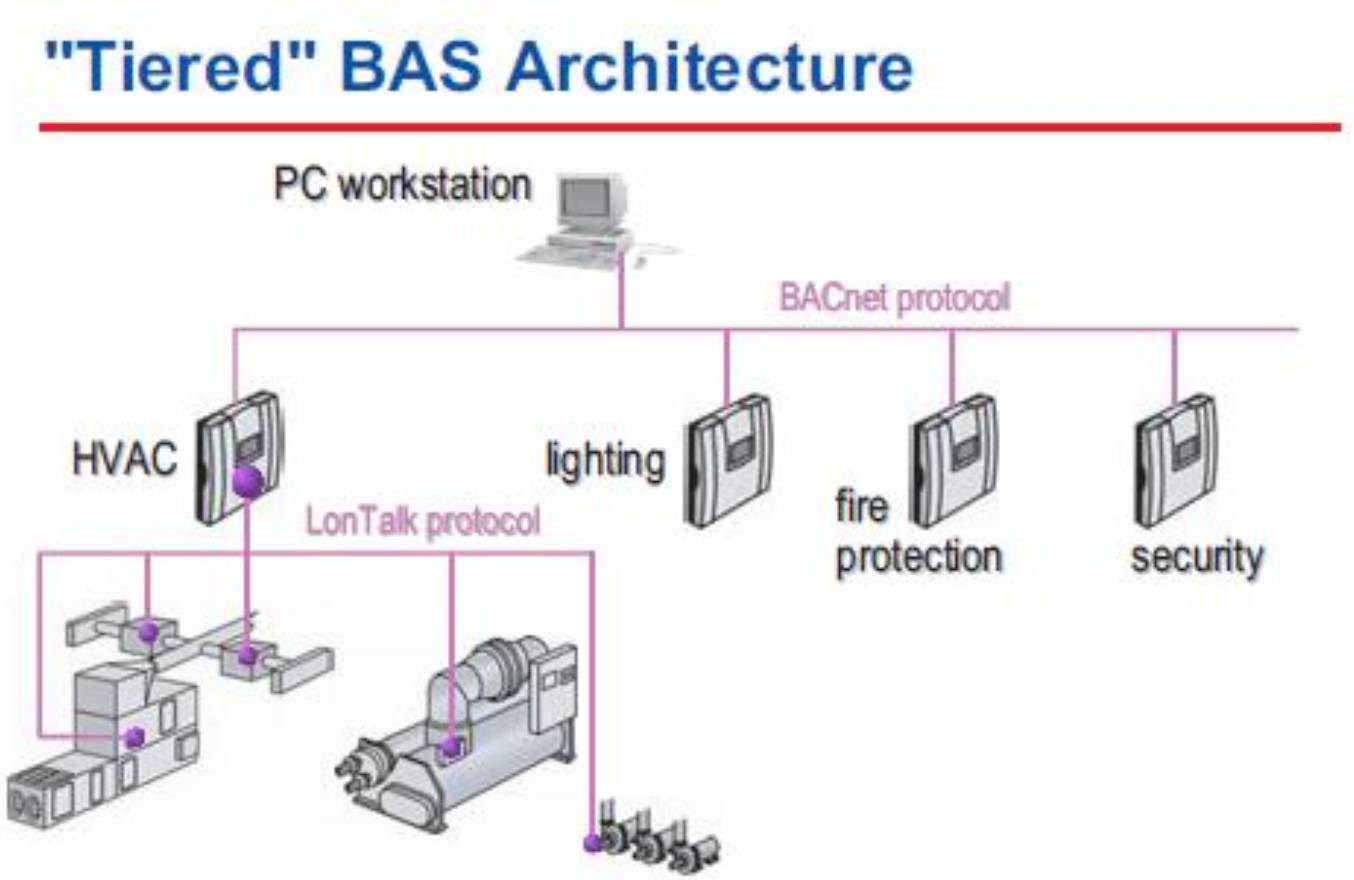
- Plug-and-play



Interoperable control systems

- Today there are dozen of communication protocols competing each other for control market.
- Two key protocols are
  - BACnet(Building automation control network) – Primarily a communication standard for integration at the data level
  - Lonworks – involves embedding neuron chips in control devices at the control layer, which will give them the power to intercommunicate as well to operate at local level

Interoperable control systems



END